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Autonomous vehicle with active wheel positioning for suspension and load manipulation

Abstract

An autonomous vehicle comprises a main body, wheels, and powered actuation systems modifying the relative positioning between each wheel and the main body. A grabbing mechanism holds a load beneath the main body. The actuation systems serve dual purposes: providing active suspension during forward motion and enabling coordinated movements to manipulate the load relative to the ground (e.g., lifting or lowering). The vehicle may be sized for sidewalk traversal and deliver consumer items, potentially in disposable packages held by the grabbing mechanism. Manipulation includes lowering and releasing the load, or positioning over, grabbing, and lifting the load. Active suspension maintains load-to-ground clearance during transport, potentially functioning as springs/dampers. Sensors provide environmental data for generating 3D representations used in controlling actuation for load manipulation. Methods include transporting the load using wheels while providing active suspension, and upon arrival, using the actuation systems for coordinated load manipulation and release.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This Application is a continuation-in-part of patent application Ser. No. 18/105,889, filed on Feb. 6, 2023, which is a continuation-in-part of patent application Ser. No. 17/246,745, filed on May 3, 2021, now U.S. Pat. No. 11,599,845, which is a continuation-in-part of patent application Ser. No. 16/502,834, filed on Jul. 3, 2019, now U.S. Pat. No. 11,046,230, which is a continuation-in-part of patent application Ser. No. 15/898,473, filed on Feb. 17, 2018, now U.S. Pat. No. 10,384,871, which claims priority to U.S. Provisional Patent Application No. 62/473,436, filed on Mar. 19, 2017.

BACKGROUND

[0002] Autonomous delivery systems are increasingly being developed to automate logistics. However, current ground-based autonomous delivery vehicles often rely on wheels, limiting their operational range to relatively smooth and level terrain. This restricts access to destinations involving stairs, curbs, or significantly uneven ground. While legged robots offer superior mobility over complex terrain, they can be less energy-efficient and slower than wheeled vehicles on flat surfaces.

SUMMARY

[0003] One embodiment is an autonomous vehicle, comprising: a main body structure; a plurality of wheels; a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure; and a grabbing mechanism coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load beneath the main body structure and above ground; wherein the autonomous vehicle is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load relative to the ground.

[0004] One embodiment is a method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; upon arrival at

the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated modification of said relative positioning between each of the wheels and the main body structure to manipulate the load relative to a ground surface; and operating the grabbing mechanism to release the load.

[0005] One embodiment is a method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; and coordinately controlling, via the control system during at least a portion of said transporting, said plurality of powered actuation systems for maintaining a clearance between a bottom surface of the load and a ground surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The embodiments are herein described by way of example only, with reference to the accompanying drawings. No attempt is made to show structural details of the embodiments in more detail than is necessary for a fundamental understanding of the embodiments. In the drawings:

[0007] FIG. 1A illustrates one embodiment of an on-road autonomous vehicle operative to autonomously collect and transport a load over public roads;

[0008] FIG. 1B illustrates one embodiment of the on-road autonomous vehicle as seen from a front view;

[0009] FIG. 1C illustrates one embodiment of the on-road autonomous vehicle as seen from a side view;

[0010] FIG. 1D illustrates one embodiment of the on-road autonomous vehicle as seen from below;

[0011] FIG. 2A illustrates one embodiment of an on-road autonomous vehicle self-driving without a load over a public road and alongside regular traffic;

[0012] FIG. 2B illustrates one embodiment of the on-road autonomous vehicle arriving at a parking lot in which a load is parked;

[0013] FIG. 2C illustrates one embodiment of the on-road autonomous vehicle getting ready to pick up the load;

[0014] FIG. 3A illustrates one embodiment of an on-road autonomous vehicle getting ready to pick up a load;

[0015] FIG. 3B illustrates one embodiment of the on-road autonomous vehicle moving/straddling over the load;

[0016] FIG. 3C illustrates one embodiment of the on-road autonomous vehicle getting into a lower position and grabbing the load;

[0017] FIG. 3D illustrates one embodiment of the on-road autonomous vehicle picking up the load;

[0018] FIG. 4A illustrates one embodiment of an on-road autonomous vehicle driving away with a load that was previously parked in a parking lot;

[0019] FIG. 4B illustrates one embodiment of the on-road autonomous vehicle self-driving with the load over a public road and alongside regular traffic;

[0020] FIG. 4C illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads;

[0021] FIG. 4D illustrates one embodiment of another method for autonomously collecting and

transporting a load over public roads;

[0022] FIG. 5A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a passenger in a cabin is located;

[0023] FIG. 5B illustrates one embodiment of the passenger in the cabin awaiting arrival of the on-road autonomous vehicle;

[0024] FIG. 5C illustrates one embodiment of the on-road autonomous vehicle picking up the cabin with the passenger;

[0025] FIG. 5D illustrates one embodiment of a method for autonomously collecting and transporting a passenger in a cabin;

[0026] FIG. 5E illustrates one embodiment of a method for requesting autonomous collection and transporting of a passenger in a cabin;

[0027] FIG. 6A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a functional load is located;

[0028] FIG. 6B illustrates one embodiment of the functional load awaiting arrival of the on-road autonomous vehicle;

[0029] FIG. 6C illustrates one embodiment of the on-road autonomous vehicle picking up and transporting the functional load;

[0030] FIG. 6D illustrates one embodiment of the functional load after being placed by the on-road autonomous vehicle at a particular location operative to work in conjunction with or support the functional load;

[0031] FIG. 6E illustrates one embodiment of the on-road autonomous vehicle driving away after placing the functional load at the particular location;

[0032] FIG. 6F illustrates one embodiment of a method for autonomously collecting transporting and placing a functional load according to a request;

[0033] FIG. 7A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a functional load is located;

[0034] FIG. 7B illustrates one embodiment of the on-road autonomous vehicle picking up the functional load and using a function associated with the functional load;

[0035] FIG. 7C illustrates one embodiment of a convoy of several on-road autonomous vehicles in which one of the on-road autonomous vehicles is carrying a functional load;

[0036] FIG. 7D illustrates one embodiment of the convoy of several on-road autonomous vehicles in which the on-road autonomous vehicles switch at least some of the loads between themselves so as to pass the functional load from one of the on-road autonomous vehicles to another of the on-road autonomous vehicles;

[0037] FIG. 7E illustrates one embodiment of the convoy of several on-road autonomous vehicles in which another of the on-road autonomous vehicles is now carrying the functional load;

[0038] FIG. 7F illustrates one embodiment of a method for autonomously collecting and using a functional load;

[0039] FIG. 7G illustrates one embodiment of a method for exchanging a functional load between at least two on-road autonomous vehicles in a convoy;

[0040] FIG. 8A illustrates one embodiment of two on-road autonomous vehicles getting into positions in conjunction with a load that is too big to be carried by only one on-road autonomous vehicle;

[0041] FIG. 8B illustrates one embodiment of the two on-road autonomous vehicles cooperatively lifting the load;

[0042] FIG. 8C illustrates one embodiment of the two on-road autonomous vehicles cooperatively lifting another load;

[0043] FIG. 8D illustrates one embodiment of a method for cooperatively lifting and transporting a load by at least two on-road autonomous vehicles;

[0044] FIG. 9A illustrates one embodiment of an on-road autonomous vehicle getting into position

behind a target vehicle;

[0045] FIG. **9B** illustrates one embodiment of the on-road autonomous vehicle now mechanically connected to the target vehicle that pulls the on-road autonomous vehicle thereby allowing for self generation of electrical energy in the on-road autonomous vehicle;

[0046] FIG. **9C** illustrates one embodiment of an on-road autonomous vehicle getting into position behind another on-road autonomous vehicle in order to be pulled thereby;

[0047] FIG. **9D** illustrates one embodiment of a method for charging batteries of an on-road autonomous vehicle on the move;

[0048] FIG. **10A** illustrates one embodiment of an on-road autonomous vehicle carrying a passenger cabin;

[0049] FIG. **10B** illustrates one embodiment of the on-road autonomous vehicle providing air gap protection to the passenger cabin;

[0050] FIG. **10C** illustrates one embodiment of the on-road autonomous vehicle providing further air gap protection to the passenger cabin;

[0051] FIG. **11A** illustrates one embodiment of an on-road autonomous vehicle about to be hit by a foreign object;

[0052] FIG. **11B** illustrates one embodiment of the on-road autonomous vehicle being hit by the foreign object;

[0053] FIG. **12A** illustrates one embodiment of an on-road autonomous vehicle carrying a load having a certain aerodynamic design and extending beyond a length of the on-road autonomous vehicle;

[0054] FIG. **12B** illustrates one embodiment of a method for adjusting an on-road autonomous vehicle to carry a long load;

[0055] FIG. **13A** illustrates one embodiment of an on-road autonomous vehicle currently integrated with a first object having drawers that are presently closed;

[0056] FIG. **13B** illustrates one embodiment of the on-road autonomous vehicle still integrated with the first object and getting one of the drawers opened;

[0057] FIG. **14A** illustrates one embodiment of an on-road autonomous vehicle currently integrated with a second object having a door that is presently closed;

[0058] FIG. **14B** illustrates one embodiment of the on-road autonomous vehicle still integrated with the second object and getting the door opened;

[0059] FIG. **15** illustrates one embodiment of a method for autonomously altering functionality of an on-road vehicle;

[0060] FIG. **16** illustrates several embodiments of on-road autonomous vehicles having several different sizes respectively;

[0061] FIG. **17A** illustrates one embodiment of a wheel robot carrying a disposable package using straps;

[0062] FIG. **17B** illustrates one embodiment of a wheel robot carrying a disposable package by inserting grabbing elements into handle holes embedded in the package;

[0063] FIG. **17C** illustrates one embodiment of a wheel robot carrying a disposable package using side arms operative to move sideways to release the package;

[0064] FIG. **17D** illustrates one embodiment of a wheel robot carrying a disposable package comprising a bag;

[0065] FIG. **18** illustrates one embodiment of a method for autonomously delivering consumer items in disposable packages;

[0066] FIG. **19A** illustrates one embodiment of an on-road autonomous vehicle carrying a first package container;

[0067] FIG. **19B** illustrates one embodiment of the on-road autonomous vehicle carrying a first package container and a second package container at the same time and in tandem;

[0068] FIG. **19C** illustrates one embodiment of the first package container comprising a plurality of

lockers containing a respective plurality of packages;

[0069] FIG. **19D** illustrates one embodiment of the second package container comprising a plurality of lockers containing a respective plurality of packages;

[0070] FIG. **19E** illustrates one embodiment of the on-road autonomous vehicle and package containers being in communicative contact with a server;

[0071] FIG. **19F** illustrates one embodiment of a route taken by the on-road autonomous vehicle in conjunction with moving and placing the package containers in different delivery locations;

[0072] FIG. **20** illustrates one embodiment of a method for autonomously delivering packages using a plurality of package containers;

[0073] FIG. **21A** illustrates one embodiment of an autonomous vehicle with a legged-wheeled hybrid locomotion system;

[0074] FIG. **21B** illustrates one embodiment of the autonomous vehicle adjusting its articulated legs to straddle over and lower its main body towards a load;

[0075] FIG. **21C** illustrates one embodiment of the autonomous vehicle transporting the load underneath its main body at a certain elevation above the ground;

[0076] FIG. **22** illustrates one embodiment of the autonomous vehicle manipulating a load onto a shelf structure;

[0077] FIG. **23** illustrates one embodiment of the autonomous vehicle navigating stairs while carrying a load;

[0078] FIG. **24** illustrates a one embodiment of a method for delivering a load autonomously using the legged-wheeled vehicle;

[0079] FIG. **25** illustrates one embodiment of a method for transporting a load autonomously while actively maintaining ground clearance using the legged-wheeled vehicle;

[0080] FIG. **26** illustrates one embodiment of a control sub-system for the autonomous vehicle;

[0081] FIG. **27** illustrates one embodiment of a method for autonomously delivering a load; and

[0082] FIG. **28** illustrates one embodiment of another method for autonomously delivering a load.

DETAILED DESCRIPTION

[0083] FIG. **1A** illustrates one embodiment of an on-road autonomous vehicle **10** operative to autonomously collect and transport a load over public roads. The on-road autonomous vehicle **10** comprises wheels **1** (**1a**, **1b**, **1c**, **1d**) touching ground **9**-ground, vertical structures **2** (**2a**, **2b**, **2c**), **2'** (**2a'**, **2b'**, **2c'**) mounted on the wheels, an upper horizontal structure **3** (**3a**, **3b**, **3c**), a plurality of sensors **4** (**4a**, **4b**, **4c**), and a first connector **5**-cnct having a certain clearance **9**-clr above ground **9**-ground.

[0084] FIG. **1B** illustrates one embodiment of the on-road autonomous vehicle **10** as seen from a front view.

[0085] FIG. **1C** illustrates one embodiment of the on-road autonomous vehicle **10** as seen from a side view.

[0086] FIG. **1D** illustrates one embodiment of the on-road autonomous vehicle **10** as seen from below. The on-road autonomous vehicle **10** further comprises a control sub-system **4** (FIG. **1A**), **6**, **7**, **8** comprising a processing unit **8** and a plurality of sensors **4** (**4a**, **4b**, **4c**, FIG. **1A**) and actuators **6**, **7**.

[0087] FIG. **2A** illustrates one embodiment of the on-road autonomous vehicle **10** self-driving without the load over a public road **20a** and alongside regular traffic **21a**, **21b**.

[0088] FIG. **2B** illustrates one embodiment of the on-road autonomous vehicle **10** arriving at a parking lot **20-p** in which the load **11** is parked. The parking lot **20-p** has a certain width **20-p-w**. **21c**, **21d**, **21e** are other vehicles, and **20b** is a public road.

[0089] FIG. **2C** illustrates one embodiment of the on-road autonomous vehicle **10** getting ready to pick up the load **11**. The load has a certain width **11-w**, and the on-road autonomous vehicle **10** has a certain width **10-w**. **23a** is a pedestrian, and **21f** is a vehicle parking alongside the load **11**.

[0090] FIG. **3A** illustrates one embodiment of the on-road autonomous vehicle **10** getting ready to

pick up the load **11**.

[0091] FIG. **3B** illustrates one embodiment of the on-road autonomous vehicle **10** moving/straddling over the load **11** and getting the connector **5-cnct** over a certain position **5-pos** associated with the load **11**.

[0092] FIG. **3C** illustrates one embodiment of the on-road autonomous vehicle **10** getting into a lower position and grabbing the load **11**.

[0093] FIG. **3D** illustrates one embodiment of the on-road autonomous vehicle **10** picking up the load **11**.

[0094] FIG. **4A** illustrates one embodiment of the on-road autonomous vehicle **10** driving away with the load **11** that was previously parked in the parking lot.

[0095] FIG. **4B** illustrates one embodiment of the on-road autonomous vehicle **10** self-driving with the load **11** over a public road **20c** and alongside regular traffic **21g**. **22a**, **22b** are traffic lights or other traffic signs. The public road **20c** has a certain lane width **20-w**. The on-road autonomous vehicle **10** has a certain width **10-w** and a certain height **10-h**.

[0096] In one embodiment, the on-road autonomous vehicle **10** is operative to autonomously collect and transport loads over public roads.

[0097] In one embodiment, the on-road autonomous vehicle comprises an upper horizontal structure **3** (**3a**, **3b**, **3c**) elevated above ground **9-ground** (FIG. **1A**) by vertical structures **2** (**2a**, **2b**, **2c**), **2'** (**2a'**, **2b'**, **2c'**) mounted on at least four wheels **1** (**1a**, **1b**, **1c**, **1d**) touching ground **9-ground**, so as to create a certain clearance **9-clr** above ground for at least a first connector **5-cnct** associated with the upper horizontal structure **3** and attached thereunder.

[0098] In one embodiment, the on-road autonomous vehicle comprises a control sub-system **4**, **6**, **7**, **8** comprising a processing unit **8** (FIG. **1D**) and a plurality of sensors **4** (**4a**, **4b**, **4c**) and actuators **6**, **7** (FIG. **1D**), in which the control sub-system is configured to generate, in real-time, a three-dimensional representation of surrounding environment using data collected by the plurality of sensors **4**.

[0099] In one embodiment, the on-road autonomous vehicle comprises at least a first linear actuator **2'+2** (**2'** moving up and down relative to **2**, i.e. **2a'** moving relative to **2a**, **2b'** moving relative to **2b**, **2c'** moving relative to **2c**, and **2d'** moving relative to **2d**) configured to control and set said certain clearance **9-clr** of the first connector **5-cnct**, by causing the first connector **5-cnct**, or the entire upper horizontal structure **3**, to move up or down relative to ground **9-ground**.

[0100] In one embodiment, the control sub-system **4**, **6**, **7**, **8** is further configured to use said three-dimensional representation, said actuators **6**, **7**, and said processing unit **8** in conjunction with a set of public-road self-driving directives, to: (i) self-drive (FIG. **2A**, FIG. **2B**) the on-road autonomous vehicle **10**, over public roads **20a** (FIG. **2A**), **20b** (FIG. **2B**) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f** (FIG. **2A**, FIG. **2B**, FIG. **2C**), to a location in which a first load **11** (FIG. **2B**) is parked, (ii) position the on-road autonomous vehicle **10** in front of the load **11** (FIG. **2C**, FIG. **3A**), and (iii) straddle the on-road autonomous vehicle **10** over the first load **11** (FIG. **3B**), such that said first connector **5-cnct** is brought to a predetermined position **5-pos** over the first load **11** (FIG. **3B**).

[0101] In one embodiment, the control sub-system **4**, **6**, **7**, **8** is further configured to use the first linear actuator **2'+2** to: (i) lower (FIG. **3C**) the first connector **5-cnct** into mechanical contact with the first load **11** thereby allowing the connector to connect to or grab the first load, and (ii) lift (FIG. **3D**) the first load **11** above ground into a position operative to self-transport (FIG. **4A**, FIG. **4B**) the first load **11** over public roads **20c** (FIG. **4B**) and alongside regular car traffic **21e** (FIG. **4A**), **21g** (FIG. **4B**).

[0102] In one embodiment, the plurality of sensors **4** comprises a plurality of digital cameras together covering front, sides, and back of the on-road autonomous vehicle **10**, thereby facilitating said generation of three-dimensional representation of surrounding environment.

[0103] In one embodiment, the plurality of sensors **4** comprises at least one radar device, thereby

farther facilitating said generation of three-dimensional representation of surrounding environment. [0104] In one embodiment, the plurality of sensors **4** comprises at least one light-detection-and-ranging (LIDAR) laser device, thereby farther facilitating said generation of three-dimensional representation of surrounding environment.

[0105] In one embodiment, the plurality of sensors **4** comprises at least two of: (i) a plurality of digital cameras, (ii) a LIDAR laser device, and (iii) a radar device, in which said generation of three-dimensional representation of surrounding environment is achieved using data fusion techniques acting on a combination of signals generated in real time by the plurality of sensors.

[0106] In one embodiment, said surrounding environment comprises at least the public roads **20a** (FIG. 2A), **20b** (FIG. 2B), **20c** (FIG. 4B) surrounding the on-road autonomous vehicle **10**, regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, **21g** surrounding the on-road autonomous vehicle **10**, traffic lights **22a**, **22b** (FIG. 4B) and other traffic signs surrounding the on-road autonomous vehicle **10**, and pedestrians **23a** (FIG. 2C) surrounding the on-road autonomous vehicle **10**, in which said three-dimensional representation in conjunction with said set of public-road self-driving directives are operative to facilitate said self-driving.

[0107] In one embodiment, said plurality of sensors **4** in conjunction with said data fusion and three-dimensional representation of surrounding environment are operative to detect said load **11** and to facilitate said positioning (FIG. 2C, FIG. 3A) of the on-road autonomous vehicle **10** in front of the load and said straddling (FIG. 3B) of the on-road autonomous vehicle **10** over the load **11**.

[0108] In one embodiment, said on-road autonomous vehicle **10** is narrow enough to facilitate said self driving over public roads **20** and in conjunction with a width **20-w** (FIG. 4B) associated with a standard lane in public roads **20c**, in which said standard lane has the width **20-w** of between 2.5 (two point five) meters and 3.7 (three point seven) meters. In one embodiment, the on-road autonomous vehicle **10** has a width **10-w** (FIG. 4B) of below 2.5 (two point five) meters. In one embodiment, said load **11** has a width **11-w** (FIG. 2C) of below 2 meters, and said load **11** is narrower than said on-road autonomous vehicle **10** (i.e., **11-w** is narrower than **10-w** in FIG. 2C), in order to facilitate said straddling (FIG. 3B) of the on-road autonomous vehicle **10** over the load **11**.

[0109] In one embodiment, said location in which the load **11** is parked is a standard parking lot **20-p** (FIG. 2B) in a parking area, in which said standard parking lot has a width **20-p-w** of between 2.3 (two point three) meters and 2.7 (two point seven) meters, and said on-road autonomous vehicle **10** is narrow enough to facilitate said straddling of the on-road autonomous vehicle over the load **11** while the load is parked in said standard parking lot **20-p** and without the on-road autonomous vehicle exceeding the width **20-p-w** of the standard parking lot (i.e., **11-w** is narrower than **10-w**, and **10-w** is narrower than **20-p-w**). In one embodiment, the on-road autonomous vehicle **10** has a width **10-w** of below 2.3 (two point three) meters. In one embodiment, said load **11** has a width **11-w** of below 1.8 (one point eight) meters, and said load **11** is narrower than said on-road autonomous vehicle **10**, in order to facilitate said straddling of the on-road autonomous vehicle **10** over the load **11** while the load is parked in said standard parking lot **20-p**.

[0110] In one embodiment, a height **10-h** (FIG. 4B) of said on-road autonomous vehicle **10** does not exceed a width **10-w** of the on-road autonomous vehicle **10**, thereby facilitating a center of gravity which is low enough to facilitate regular car traffic maneuvers in conjunction with said public roads **20a** (FIG. 2A), **20b** (FIG. 2B), **20c** (FIG. 4B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, **21g**. In one embodiment, said maneuvers comprise traveling at a velocity exceeding 80 (eighty) kilometers-per-hour. In one embodiment, said maneuvers comprise a centripetal acceleration of above 3 (three) meters-per-second-square (m/s*s). In one embodiment, said width **10-w** is between 2 (two) meters and 2.5 (two point five) meters, and said height **10-h** is below 2 (two) meters.

[0111] In one embodiment, said actuators **6**, **7** comprise electrical motors **6b**, **6d** capable of linearly accelerating and de-accelerating the on-road autonomous vehicle **10** at a rate of at least 3 (three) meters-per-second-square (m/s*s), thereby avoiding car-accidents in conjunction with said public

roads **20a** (FIG. 2A), **20b** (FIG. 2B), **20c** (FIG. 4B) and regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, **21g**.

[0112] In one embodiment, said lifting (FIG. 3D) is done so as to lift the load **11** to a position which is not more than 50 cm (fifty centimeters) above ground, thereby facilitating a center of gravity which is low enough to facilitate regular car traffic maneuvers in conjunction with said public roads **20a** (FIG. 2A), **20b** (FIG. 2B), **20c** (FIG. 4B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, **21g**.

[0113] In one embodiment, said positioning and straddling comprises: using said three-dimensional representation of surrounding environment to accurately represent the load **11** and position the on-road autonomous vehicle **10** in front to the load, and using slow and controlled acceleration to slowly move forward while maintaining constant and regular spacing between the load **11** and the inner surfaces of the on-road autonomous vehicle **10**, until predicting accurate positioning of the on-road autonomous vehicle **10** above the load **11**.

[0114] In one embodiment, said first liner actuator **2'+2** is a distributed linear actuator comprising several sub-actuators (e.g., four sub-actuators: **2a'** moving relative to **2a**, **2b'** moving relative to **2b**, **2c'** moving relative to **2c**, and **2d'** moving relative to **2d**), in which each sub-actuator is associated with one of the wheels, such that the entire upper horizontal structure **3** is operative to move up and down relative to the wheels and ground.

[0115] In one embodiment, the sub-actuators are also operative to act as springs or mechanical dampers for the wheels relative to the upper horizontal structure.

[0116] In one embodiment, said first liner actuator is embedded in the first connector **5-cnct**, thereby causing only the connector to move up and down relative to ground, and such that the upper horizontal structure **3** remains in place.

[0117] FIG. 4C illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads. In step **1001**, receiving, in conjunction with an on-road autonomous vehicle **10**, a request to collect-and-transport a load **11** which is currently parked in a certain location **20-p**. In step **1002**, self-driving (FIG. 2A, FIG. 2B, FIG. 2C), by the on-road autonomous vehicle, over public roads **20a** (FIG. 2A), **20b** (FIG. 2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, from a current location of the on-road autonomous vehicle to said certain location **20-p** of the load **11**. In step **1003**, upon arrival to said certain location **20-p**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle **10**, over the load **11**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the load **11** above ground in a linear upward movement that creates a full clearance of the load **11** above ground. In step **1004**, transporting autonomously (FIG. 4A, FIG. 4B) the load **11**, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21e** (FIG. 4A), **21g** (FIG. 4B), while the load **11** is hanging underneath the on-road autonomous vehicle **10** and such that the entire load **11** maintains said full clearance above ground during transport. In one embodiment, autonomously navigating, by the on-road autonomous vehicle, to a destination location, and lowering the load **11** at the destination location in a linear downward movement that places the load on the ground (a reverse transition from FIG. 3D to FIG. 3C).

[0118] FIG. 4D illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads. In step **1011**, self-driving using a set of public-road self-driving directives, by an on-road autonomous vehicle **10**, over public roads **20a** (FIG. 2A), **20b** (FIG. 2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, from a current location of the on-road autonomous vehicle to a certain location **20-p** of the load **11**. In step **1012**, upon arrival to said certain location **20-p**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), using a set of autonomous straddling directives, by the on-road autonomous vehicle **10**, over the load **11**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the load **11** above ground in a

linear upward movement that creates a full clearance of the load **11** above ground. In step **1013**, transporting autonomously (FIG. 4A, FIG. 4B) the load **11** using a set of public-road autonomous transport directives, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21e** (FIG. 4A), **21g** (FIG. 4B), while the load **11** is hanging underneath the on-road autonomous vehicle **10** and such that the entire load **11** maintains said full clearance above ground during transport. In one embodiment, said public-road self-driving directives are fine-tuned to facilitate said self-driving during a period that the on-road autonomous vehicle **10** does not transport the load **11**, and is therefore (i) lighter and (ii) has a higher center of gravity; said public-road autonomous transport directives are fine-tuned to facilitate said transporting during a period that the on-road autonomous vehicle **10** transports the load **11**, and is therefore (i) heavier and (ii) has a lower center of gravity; and said autonomous straddling directives are fine-tuned to facilitate said straddling by directing a slow approach and slow straddling of the on-road autonomous vehicle **10** over the load **11** and into an accurate final position above the load.

[0119] FIG. 5A illustrates one embodiment of an on-road autonomous vehicle **10** self-driving to a location in which a passenger in a cabin is located.

[0120] FIG. 5B illustrates one embodiment of the passenger **30** in the cabin **11** awaiting arrival of the on-road autonomous vehicle **10**. **5-pos** is a position in the cabin **11** that facilitates grabbing or connecting to the cabin **11**.

[0121] FIG. 5C illustrates one embodiment of the on-road autonomous vehicle **10** picking up the cabin **11** with the passenger **30**. The cabin **11** is now connected to the on-road autonomous vehicle **10** via connector **5-cnct** and in conjunction with position **5-pos**.

[0122] FIG. 5D illustrates one embodiment of a method for autonomously collecting and transporting a passenger in a cabin. In step **1021**, receiving, in conjunction with an on-road autonomous vehicle **10** (FIG. 5A), from a passenger **30** (FIG. 5B) associated with a cabin **11** (FIG. 5B) located in a certain location, a request to collect-and-transport the passenger together with the cabin. In step **1022**, self-driving (FIG. 2A), by the on-road autonomous vehicle **10**, from a current location of the on-road autonomous vehicle to said certain location of the cabin **11** (FIG. 5B). In step **1023**, upon arrival to said certain location: (i) confirming that the passenger **30** is indeed in the cabin **11** (FIG. 5B), and (ii) straddling autonomously, by the on-road autonomous vehicle **10**, over the cabin **11** (FIG. 3B), thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously (FIG. 3C, FIG. 3D) the cabin **11**, together with the passenger **30**, above ground (FIG. 5C). In step **1024**, transporting autonomously the cabin **11** together with the passenger **30**, by the on-road autonomous vehicle **10**, while the cabin **11** is hanging underneath the on-road autonomous vehicle (FIG. 4B). In one embodiment, the method further includes: receiving, in conjunction with the on-road autonomous vehicle **10**, a second request to collect-and-transport a cargo **11** which is currently located in a second location **20-p** (FIG. 2B); self-driving (FIG. 2A, FIG. 2B, FIG. 2C), by the on-road autonomous vehicle **10**, over public roads **20a** (FIG. 2A), **20b** (FIG. 2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, from a current location of the on-road autonomous vehicle to said second location **20-p**; upon arrival to said second location **20-p**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle **10**, over the cargo **11**, thereby allowing the on-road autonomous vehicle **10** to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the cargo **11** above ground in a linear upward movement that creates a full clearance of the cargo **11** above ground; and transporting autonomously (FIG. 4A, FIG. 4B) the cargo **11**, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21e** (FIG. 4A), **21g** (FIG. 4B), while the cargo **11** is hanging underneath the on-road autonomous vehicle **10** and such that the entire cargo **11** maintains said full clearance above ground during transport, thereby facilitating dual use of the on-road autonomous vehicle **10** for both said transporting of the passenger **30** during a certain period of time and said transporting of the cargo **11** during another period of time. In one embodiment, said transporting of the passenger **30** or other passengers is done during the

mornings or the evenings and in conjunction with work rush hours, while said transporting of the cargo **11** or other cargo is done during mid-day hours, while most people are working.

[0123] FIG. 5E illustrates one embodiment of a method for requesting autonomous collection and transporting of a passenger in a cabin. In step **1031**, associating a cabin **11** with a passenger **30** (FIG. 5B). In step **1032**, sending, in conjunction with the cabin **11**, a request to collect-and-transport the cabin together with the passenger **30**. In step **1033**, receiving, in conjunction with the cabin **11**, a ready-to-collect message from an on-road autonomous vehicle **10** (FIG. 5A). In step **1034**, sending, in conjunction with the cabin **11**, a ready-to-be-collected message to the on-road autonomous vehicle **10**, thereby facilitating collection and transporting of the passenger **30** in the cabin **11** (FIG. 5C).

[0124] FIG. 6A illustrates one embodiment of an on-road autonomous vehicle **10** self-driving to a location **28** (FIG. 6B) in which a functional load **11** is located.

[0125] FIG. 6B illustrates one embodiment of the functional load **11** awaiting arrival of the on-road autonomous vehicle **10**.

[0126] FIG. 6C illustrates one embodiment of the on-road autonomous vehicle **10** picking up and transporting the functional load **11**.

[0127] FIG. 6D illustrates one embodiment of the functional load **11** after being placed by the on-road autonomous vehicle **10** at a particular location **29** operative to work in conjunction with or support the functional load **11** using an interface **29-rec**.

[0128] FIG. 6E illustrates one embodiment of the on-road autonomous vehicle **10** driving away after placing the functional load **11** at the particular location.

[0129] FIG. 6F illustrates one embodiment of a method for autonomously collecting transporting and placing a functional load according to a request. In step **1041**, receiving, in conjunction with an on-road autonomous vehicle **10** (FIG. 6A), a request to place a specific load **11** (FIG. 6B) at a particular location **29** (FIG. 6D), in which the specific load **11** is operative to perform a specific function. In step **1042**, self-driving, by the on-road autonomous vehicle **10**, to a specific storage location **28** (FIG. 6B) of the specific load **11**. In step **1043**, upon arrival to said specific storage location **28**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle **10**, over the specific load **11**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the specific load **11** above ground in a linear upward movement that creates a full clearance of the specific load **11** above ground. In step **1044**, transporting autonomously (FIG. 4B) the specific load **11**, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21g** (FIG. 4B), to the particular location **29** (FIG. 6D), while the specific load **11** is hanging (FIG. 6C) underneath the on-road autonomous vehicle **10** and such that the entire specific load **11** maintains said full clearance above ground during transport. In step **1045**, upon arrival to the particular location **29**, lowering the specific load **11** at the particular location **29** (FIG. 6D) in a linear downward movement that places the load down (a reverse transition from FIG. 3D to FIG. 3C), thereby facilitating performance of the specific action in conjunction with the particular location **29**.

[0130] In one embodiment, said linear downward movement creates a physical contact between the specific load **11** and a reception element or interface **29-rec** (FIG. 6D) at the particular location **29**, thereby enabling said specific function in conjunction with the physical contact.

[0131] In one embodiment, the specific function is fluid or gas transfer, and the physical contact with the interface **29-rec**, which comprises ducts, enables said fluid or gas transfer.

[0132] In one embodiment, the specific function is electrical charge transfer, and the physical contact with the interface **29-rec**, which comprises electrical contact, enables said electrical charge transfer.

[0133] In one embodiment, said specific function comprises at least one of: (i) fluid transfer such as water or gasoline transfer, (ii) gas transfer such as methane transfer, (iii) electrical charge

transfer in which the specific load **11** is a rechargeable battery, (iv) automatic vending in which the specific load **11** is an automatic vending machine, (v) communication relaying in which the specific load **11** is a communication relay or a cellular base station, (vi) waste collection in which the specific load **11** is a waste container, (vii) monitoring, surveillance, or intelligence gathering, and (viii) distribution of objects such as drones, other autonomous vehicles, and munitions.

[0134] One embodiment further comprises straddling away (FIG. 6E) from the specific load **11**, by the on-road autonomous vehicle **10**, thereby leaving the specific load **11** (FIG. 6D) at the particular location **29** to perform said specific function.

[0135] FIG. 7A illustrates one embodiment of an on-road autonomous vehicle **10** self-driving to a location **27** in which a functional load **11-func** is located.

[0136] FIG. 7B illustrates one embodiment of the on-road autonomous vehicle **10** picking up the functional load **11-func**, interfacing **5-int** with the functional load **11-func**, and using a function associated with the functional load **11-func**.

[0137] FIG. 7C illustrates one embodiment of a convoy of several on-road autonomous vehicles **10a**, **10b**, **10c**, **10d** in which one of the on-road autonomous vehicles **10a** is carrying a functional load **10-func**. **11b**, **11c**, **11d** are general loads.

[0138] FIG. 7D illustrates one embodiment of the convoy of several on-road autonomous vehicles **10a**, **10b**, **10c**, **10d** in which the on-road autonomous vehicles switch at least some of the loads between themselves so as to pass the functional load **11-func** from one of the on-road autonomous vehicles **10a** to another **10b** of the on-road autonomous vehicles.

[0139] FIG. 7E illustrates one embodiment of the convoy of several on-road autonomous vehicles **10a**, **10b**, **10c**, **10d** in which another of the on-road autonomous vehicles **10b** is now carrying the functional load **11-func**.

[0140] FIG. 7F illustrates one embodiment of a method for autonomously collecting and using a functional load. In step **1051**, determining, by an on-road autonomous vehicle **10** (FIG. 7A), that a certain function is needed in conjunction with operating said on-road autonomous vehicle. In step **1052**, self-driving (FIG. 2A), by the on-road autonomous vehicle **10**, from a current location of the on-road autonomous vehicle to a certain location **27** (FIG. 7A) where a functional load **11-func** is parked, in which said functional load is operative to render said certain function needed. In step **1053**, upon arrival to said certain location (FIG. 7A), straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle **10**, over the functional load **10-func**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the functional load **10-func** above ground in a linear upward movement that creates a full clearance of the load above ground (the result is illustrated in FIG. 7B). In step **1054**, interfacing **5-int** (FIG. 7B), in a physical manner, between the on-road autonomous vehicle **10** and the functional load **11-func**, thereby facilitating said rendering of the certain function needed from the functional load to the on-road autonomous vehicle.

[0141] In one embodiment, said certain function needed is a need to charge an electrical battery **12a**, **12c** (FIG. 7B) belonging to the on-road autonomous vehicle **10**, said functional load **11-func** is an energy source, and said interfacing **5-int** is operative to transfer energy from the functional load **11-func** to the on-road autonomous vehicle **10**. In one embodiment, said energy source **11-func** is a portable battery, and said interfacing **5-int** is an electrical interface operative to transport electricity from the portable battery **11-func** to the battery **12a**, **12c** of the on-road autonomous vehicle **10**. In one embodiment, said energy source **11-func** is a portable fuel cell, and said interfacing **5-int** is an electrical interface operative to transport electricity from the fuel cell **11-func** to the battery **12a**, **12c** of the on-road autonomous vehicle **10**. In one embodiment, said energy source **11-func** is a portable generator with on-board fuel, and said interfacing **5-int** is an electrical interface operative to transport electricity from the portable generator **11-func** to the battery **12a**, **12c** of the on-road autonomous vehicle **10**.

[0142] In one embodiment, said certain function needed is a need for fuel, said functional load **11-func** is a fuel tank, and said interfacing **5-int** is operative to transfer fuel from the fuel tank **11-func** to the on-road autonomous vehicle **10**. In one embodiment, said fuel is gasoline or diesel fuel. In one embodiment, said fuel is a fuel operative to drive a fuel cell, such as hydrogen fuel, methanol fuel, ethanol fuel, or methane fuel.

[0143] FIG. 7G illustrates one embodiment of a method for exchanging a functional load between at least two on-road autonomous vehicles in a convoy. In step **1061**, detecting, in a moving convoy (FIG. 7C) comprising on-road autonomous vehicles **10a**, **10b**, **10c**, **10d**, a need of one of the on-road autonomous vehicles **10b** in the convoy to use a certain function, and further detecting another on-road autonomous vehicle **10a** in the moving convoy that currently makes use of said certain function in conjunction with a functional load **11-func** carried therewith (i.e., **10a** carries **11-func** for a certain use, and **10b** currently needs **11-func** for that use). In step **1062**, stopping the convoy (FIG. 7D) as a response to said detections. In step **1063**, autonomously offloading, by said another on-road autonomous vehicle **10a**, the functional load **11-func**, thereby placing the functional load **11-func** on ground (e.g., in FIG. 7D, **11-func** has now been placed on ground, and **10a** which previously carried **11-func** has now moved to the back of the convoy). In step **1064**, autonomously picking-up the functional load **11-func** off ground and interfacing to said functional load by said one of the on-road autonomous vehicles **10b** (e.g., FIG. 7E, **10b** has now collected **11-func**, and load **11b** which was previously carried by **10b** is now carried by yet another vehicle **10c**), thereby facilitating said certain function in conjunction with said one of the on-road autonomous vehicles **10b**.

[0144] In one embodiment, said picking-up of the functional load **11-func**, by said one of the on-road autonomous vehicles **11b** (FIG. 7E), is facilitated by straddling autonomously, by said one of the on-road autonomous vehicle **11b**, over the functional load **11-func** now on ground (FIG. 7D, **10b** is driving/straddling over **11-func**), thereby allowing said one of the on-road autonomous vehicle **10b** to grab and lift autonomously the functional load **11-func** above ground in a linear upward movement that creates a full clearance of the functional load above ground (in FIG. 7E, **10b** has lifted **11-func** above ground), thereby facilitating said interfacing.

[0145] In one embodiment, said certain function is an electrical charging of batteries **12a**, **12c** belonging to said one of the on-road autonomous vehicles **10b**, in which said functional load **11-func** in an energy source.

[0146] One embodiment further comprising resuming movement by the convoy.

[0147] FIG. 8A illustrates one embodiment of two on-road autonomous vehicles **10a**, **10b** getting into positions in conjunction with a load **11-long** that is too big to be carried by only one on-road autonomous vehicle. **5-cnct-1** is a connector of on-road autonomous vehicles **10a** and is operative to grab a grabbing point **5-grb-1** of load **11-long** at position **5-pos-1**. **5-cnct-2** is a connector of on-road autonomous vehicles **10b** and is operative to grab a grabbing point **5-grb-2** of load **11-long** at position **5-pos-2**.

[0148] FIG. 8B illustrates one embodiment of the two on-road autonomous vehicles **10a**, **10b** cooperatively lifting the load **11-long**.

[0149] FIG. 8C illustrates one embodiment of the two on-road autonomous vehicles **10a**, **10b** cooperatively lifting another load **11-wide**. **11-ext-1** and **11-ext-2** are extensions of load **11-wide**.

[0150] FIG. 8D illustrates one embodiment of a method for cooperatively lifting and transporting a load by at least two on-road autonomous vehicles. In step **1071**, driving autonomously, by a first on-road autonomous vehicle **10a** (FIG. 8A), into a first predetermined position **5-pos-1** over a load **11-long** (FIG. 8A), in which the first predetermined position **5-pos-1** is associated with a first grabbing point **5-grb-1** in the load **11-long**. In step **1072**, driving autonomously, by a second on-road autonomous vehicle **10b** (FIG. 8A), into a second predetermined position **5-pos-2** over the load **11-long**, in which the second predetermined position **5-pos-2** is associated with a second grabbing point **5-grb-2** in the load **11-long**. In step **1073**, lifting the load **11-long** together and

synchronously (FIG. 8B), by the first and the second on-road autonomous vehicles **10a**, **10b** in conjunction respectively with the first and second grabbing points **5-grb-1**, **5-grb-2**, such that: (i) the load **11-long** is lifted above ground, (ii) the load **11-long** achieves full clearance above ground, and (iii) a weight of the load **11-long** is spread between the first **10a** and second **10b** on-road autonomous vehicles via the first **5-grb-1** and second **5-grb-2** grabbing points respectively. In step **1074**, transporting autonomously the load **11-long** together and synchronously by the first and second on-road autonomous vehicles **10a**, **10b**, such that the load **11-long** maintains said full clearance above ground during transport.

[0151] In one embodiment, the first on-road autonomous vehicle **10a** constantly communicates with the second **10b** on-road autonomous vehicle during said lifting and transport in order to achieve said synchronicity. In one embodiment, the first on-road autonomous vehicle **10a** controls the second on-road autonomous vehicle **10b** during said lifting and transport, thereby facilitating the lifting and transport autonomously. In one embodiment, the first on-road autonomous vehicle **10a** receives sensory input from the second on-road autonomous vehicle **10b** during said lifting and transport, thereby facilitating the lifting and transport autonomously.

[0152] In one embodiment, said driving autonomously comprises straddling autonomously over the load, by said first and second on-road autonomous vehicles **10a**, **10b**.

[0153] In one embodiment, said lifting comprises: lowering a first connector **5-cnct-1** of the first on-road autonomous vehicle **10a** into mechanical contact with the first grabbing point **5-grb-1**, lowering a second connector **5-cnct-2** of the second on-road autonomous **10b** vehicle into mechanical contact with the second grabbing point **5-grb-2**, grabbing the first grabbing point **5-grb-1** by the first connector **5-cnct-1**, grabbing the second grabbing point **5-grb-2** by the second connector **5-cnct-2**, and raising synchronously the first and second connectors **5-cnct-1**, **5-cnct-2** respectively by the first and second on-road autonomous vehicles **10a**, **10b**.

[0154] In one embodiment, the load **11-long** (FIG. 8A, FIG. 8B) is narrower than the on-road autonomous vehicles **10a**, **10b**, thereby enabling the entire load to be carried under the on-road autonomous vehicles.

[0155] In one embodiment, the load **11-wide** (FIG. 8C) is wider than the on-road autonomous vehicles **10a**, **10b**, thereby requiring the load to be supported under the on-road autonomous vehicles by two narrow extension shafts **11-ext-1**, **11-ext-2** (FIG. 8C).

[0156] FIG. 9A illustrates one embodiment of an on-road autonomous vehicle **10** getting into position behind a target vehicle **10-target**. **10-mech** is a mechanical hook or grabbing mechanism.

[0157] FIG. 9B illustrates one embodiment of the on-road autonomous vehicle **10** now mechanically connected via the mechanical hook or grabbing mechanism **10-mech** to the target vehicle **10-target**. The target vehicle **10-target** pulls the on-road autonomous vehicle thereby allowing the on-road autonomous vehicle to self generate electrical energy.

[0158] FIG. 9C illustrates one embodiment of an on-road autonomous vehicle **10a** getting into position behind another on-road autonomous vehicle **10b** in order to be pulled thereby.

[0159] FIG. 9D illustrates one embodiment of a method for charging batteries of an on-road autonomous vehicle on the move. In step **1081**, identifying, by an on-road autonomous vehicle **10** (FIG. 9A), a target vehicle **10-target** (FIG. 9A) operative to mechanically pull another vehicle. In step **1082**, self driving (FIG. 2A), by the on-road autonomous vehicle **10**, into a position behind the target vehicle identified (FIG. 9A). In step **1083**, connecting autonomously (FIG. 9B), by the on-road autonomous vehicle **10**, to the target vehicle **10-target**, by performing an autonomous maneuver in conjunction with a mechanical hook or grabbing mechanism **10-mech**, so as to mechanically connect between the on-road autonomous vehicle **10** and the target vehicle **10-target** (FIG. 9B). In step **1084**, being pulled, by the target vehicle **10-target**, thereby causing at least one wheel **1d** (FIG. 9B) of the on-road autonomous vehicle **10** to rotate as the on-road autonomous vehicle **10** is pulled by the target vehicle **10-target**. In step **1085**, generating electrical energy from at least one dynamo **6d** (FIG. 9B) connected to the at least one wheel **1d** now rotating, thereby

allowing the dynamo **6d** to charge a battery **12c** of the on-road autonomous vehicle **10**.

[0160] In one embodiment, said connecting autonomously is done during forward movement of both the on-road autonomous vehicle **10** and the target vehicle **10-target**.

[0161] In one embodiment, said connecting autonomously is done in a stationary state, before forward movement of both the on-road autonomous vehicle **10** and the target vehicle **10-target**, in which said forward movement by the target vehicle causes said pulling.

[0162] In one embodiment, said mechanical hook or grabbing mechanism **10-mech** is a part of the on-road autonomous vehicle **10**. In one embodiment, the hook or grabbing mechanism **10-mech** is a moving hook. In one embodiment, the hook or grabbing mechanism **10-mech** is a mechanical connector.

[0163] In one embodiment, the target vehicle **10-target** is a second on-road autonomous vehicle **10b** (FIG. **9C**), in which the second on-road autonomous vehicle **10b** synchronizes said autonomous connection with the on-road autonomous vehicle **10** or **10a**. In one embodiment, said self driving, by the on-road autonomous vehicle **10** or **10a**, into a position behind the target vehicle **10-target**, is done in conjunction with the second on-road autonomous vehicle **10b**, **10-target** also self driving into a position in front of the on-road autonomous vehicle **10** or **10a**.

[0164] In one embodiment, the dynamo **6d** is an electrical engine of the on-road autonomous vehicle **10**, in which the electrical engine is operated in a breaking mode, thereby facilitating said generation of electrical energy.

[0165] In one embodiment, said identification is done so as to identify the target vehicle **10-target** as a vehicle currently traveling in a direction similar to a direction desirable by the on-road autonomous vehicle **10**.

[0166] FIG. **10A** illustrates one embodiment of an on-road autonomous vehicle **10** carrying a passenger cabin **11**.

[0167] FIG. **10B** illustrates one embodiment of the on-road autonomous vehicle **10** providing air gap protection **11-10-gap-1**, **11-10-gap-2** to the passenger cabin **11**.

[0168] FIG. **10C** illustrates one embodiment of the on-road autonomous vehicle **10** providing further air gap protection **11-10-gap-3**, **11-10-gap-4**, **11-10-gap-5**, **11-10-gap-6** to the passenger cabin **11**.

[0169] One embodiment is a system operative to provide a hybrid air-gap and mechanical protection for a passenger cabin. The system includes a passenger cabin **11** (FIG. **10A**), and an on-road autonomous vehicle **10** (FIG. **10A**) operative to straddle over a passenger cabin **11** and then to pick up the passenger cabin **11**, such that the passenger cabin **11** is carried underneath the on-road autonomous vehicle **10** (FIG. **10A**). The on-road autonomous vehicle **10** is operative to mechanically enclose the passenger cabin **11** from at least 4 (four) directions, such as to provide mechanical protection from impact with foreign objects, in which said mechanical protection is enhanced by maintaining air-gaps **11-10-gap-1,2,3,4,5,6** between the on-road autonomous vehicle **10** and the passenger cabin **11** in conjunction with the at least 4 (four) directions.

[0170] In one embodiment, said at least 4 (four) directions are front, rear, left, and right, in which the front direction is associated with one of the air-gaps **10-11-gap-5** (FIG. **10C**) located in front of the passenger cabin **11**, the rear direction is associated with one of the air-gaps **10-11-gap-3** (FIG. **10C**) located behind the passenger cabin **11**, the left direction is associated with one of the air-gaps **10-11-gap-1** (FIG. **10B**) located to the left of the passenger cabin **11**, and the right direction is associated with one of the air-gaps **10-11-gap-2** (FIG. **10B**) located to the right of the passenger cabin **11**. In one embodiment, said at least 4 (four) directions are at least 5 (five) directions comprising also an up direction associated with one of the air-gaps **10-11-gap-6** (FIG. **10C**) located above the passenger cabin **11**. In one embodiment, said at least 5 (five) directions are 6 (directions) directions comprising also a down direction associated with one of the air-gaps **10-11-gap-4** (FIG. **10C**) located below the passenger cabin **11**.

[0171] FIG. **11A** illustrates one embodiment of an on-road autonomous vehicle **10** about to be hit

by a foreign object **21**. The on-road autonomous vehicle **10** comprises a rear section **10-rear**, a front section **10-front**, and a piston or spring **10-sp**.

[0172] FIG. **11B** illustrates one embodiment of the on-road autonomous vehicle **10** being hit by the foreign object **21h**.

[0173] One embodiment is a system operative to protect an on-road vehicle from impact with foreign objects. The system includes a front section **10-front** (FIG. **11A**) of an on-road vehicle **10** (FIG. **11A**), in which the front section comprises two front wheels **1c**, **1d** on which the on-road vehicle is supported. The system further includes a rear section **10-rear** (FIG. **11A**) of the on-road vehicle **10**, in which the rear section comprises two rear wheels **1a**, **1b** on which the on-road vehicle is further supported, and in which the front section **10-front** is mechanically connected to the rear section **10-rear** so as to allow movement of the entire front section **10-front** relative to the rear sections **10-rear**. The system further includes a passenger or cargo cabin **11** of the on-road vehicle **10**, in which the passenger or cargo cabin is mechanically connected to the rear section **10-rear** without touching the front section **10-front**. The system further includes a linear horizontal actuator or horizontal spring **10-sp** (FIG. **11A**) having two sides, in which the linear horizontal actuator or horizontal spring is mechanically connected to the front section **10-front** on one side and to the rear section **10-rear** on the other side, and in which the linear horizontal actuator or horizontal spring **10-sp** is operative to generate a reaction force pushing the front section **10-front** away from the rear section **10-rear** when the front section moves toward the rear section. During a collision of the front section **10-front** with a foreign object **21h** (FIG. **11B**), the front section **10-front** moves toward the rear section **10-rear** (FIG. **11B**), thereby causing said reaction force to accelerate the rear section **10-rear** away from the foreign object **21h**, and thereby avoiding a collision between the passenger or cargo cabin **11** and the foreign object **21h** or at least reducing a relative velocity between the passenger or cargo cabin **11** and the foreign object **21h**. In one embodiment, the linear horizontal actuator or horizontal spring is a piston.

[0174] One embodiment is an on-road vehicle operative to control a length thereof. The on-road vehicle includes a front section **10-front** (FIG. **11A**) of the on-road vehicle **10** (FIG. **11A**), in which the front section comprises two front wheels **1c**, **1d** on which the on-road vehicle is supported. The on-road vehicle further includes a rear section **10-rear** (FIG. **11A**) of the on-road vehicle **10**, in which the rear section comprises two rear wheels **1a**, **1b** on which the on-road vehicle is further supported, and in which the front section **10-front** is mechanically connected to the rear section **10-rear** so as to allow movement of the entire front section **10-front** relative to the rear sections **10-rear**. The on-road vehicle further includes a linear horizontal actuator or a piston **10-sp** (FIG. **11A**) having two sides, in which the linear horizontal actuator or piston is mechanically connected to the front section **10-front** on one side and to the rear section **10-rear** on the other side, and in which the linear horizontal actuator or piston **10-sp** is operative to adjust a distance between the front section **10-front** and the rear section **10-rear** so as to control a length of the on-road vehicle.

[0175] In one embodiment, said length is adjusted to support loads **11** of different lengths to be carried by the on-road vehicle.

[0176] In one embodiment, said length is adjusted to support compact length during parking of the on-road vehicle **10**.

[0177] In one embodiment, said length is adjusted to support extended length during high speed driving of the on-road vehicle **10** in order to protect a load **11** carried by the on-road vehicle by increasing an air-gap **11-10-gap-5** (FIG. **10C**) between the load **11** the front section **10-front**.

[0178] FIG. **12A** illustrates one embodiment of an on-road autonomous vehicle **10** carrying a load **11** having a certain aerodynamic design and extending beyond a length of the on-road autonomous vehicle.

[0179] One embodiment is a combined vehicle-and-load arrangement operative to reduce drag on the vehicle. The combined vehicle-and-load arrangement includes an on-road autonomous vehicle **10** (FIG. **12A**) having a first aerodynamic drag coefficient and a load **11** (FIG. **12A**) having a

certain aerodynamic design. The on-road autonomous vehicle **10** is configured to straddle autonomously over the load **11**, thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously the load **11** above ground in a linear upward movement that creates a full clearance of the load above ground. The combined vehicle-and-load arrangement **10+11** of the on-road autonomous vehicle together with the load now lifted is operative to reduce the first aerodynamic drag coefficient of the on-road autonomous vehicle.

[0180] FIG. **12B** illustrates one embodiment of a method for adjusting an on-road autonomous vehicle to carry a long load. In step **1091**, straddling autonomously, by an on-road autonomous vehicle **10** having a certain length **10-L** (FIG. **12**), over a load **11** having a greater length **11-L** (FIG. **12**), thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously the load **11** above ground in a linear upward movement that creates a full clearance of the load above ground, and resulting in the load **11** extending beyond said certain length **10-L**. In step **1092**, detecting, by sensors **4a**, **4b**, **4c** (FIG. **12**) belonging to the on-road autonomous vehicle **10**, said extension of the load **11** beyond said certain length **10-L**. In step **1093**, altering, according to said detection, at least a first parameter in conjunction with an autonomous driving procedure of the on-road autonomous vehicle, thereby adapting the on-road autonomous vehicle **10** to self-drive with the load **11** extending beyond said certain length **10-L**.

[0181] In one embodiment, the first parameter is an effective length of the on-road autonomous vehicle, in which said effective length is increased from the certain length **10-L** to the greater length **11-L**.

[0182] One embodiment is a system operative to autonomously collect and transport a passenger **30** in a cabin **11** (FIG. **5C**), comprising: an autonomous on-road vehicle **10** operative to straddle over (FIG. **3B**, and the transition from FIG. **3A** to FIG. **3B**) loads **11** such as cabins operative to contain passengers **30** (FIG. **5C**); and a cabin **11** operative to contain passengers **30** (FIG. **5B**), in which the cabin **11** is currently located at a certain parking location without any passengers inside (the cabin **11** as shown in FIG. **5B**, but without the passenger **30** inside); wherein, the system is configured to: receive a request to collect-and-transport a passenger **30** which is currently located or is soon-to-be-located in a pick-up location; self-drive, as a response to said request, the on-road autonomous vehicle **10**, to said certain parking location; upon arrival of the on-road autonomous vehicle **10** to said certain parking location: straddle autonomously (FIG. **3B**, and the transition from FIG. **3A** to FIG. **3B**), the on-road autonomous vehicle **10**, over the cabin **11**, thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously (the transition between FIG. **3C** and FIG. **3D**) the cabin **11**; self-drive the on-road autonomous vehicle **10** to said pick-up location, while the cabin **11** is hanging underneath the on-road autonomous vehicle; and pick-up the passenger **30** (FIG. **5C**) at the pick-up location.

[0183] One embodiment is a system operative to transport passengers in a cabin and also transport cargo loads, comprising: an autonomous on-road vehicle **10** operative to straddle over (FIG. **3A**, FIG. **3B**) loads **11** such as cabins operative to contain passengers and such as cargo loads; a cabin **11** operative to contain passengers **30** (FIG. **5C**), in which the cabin is currently located at a certain parking location; and a cargo load **11** (FIG. **2C**); wherein, the system is configured to: autonomously transport the cargo load **11** (FIG. **4B**), using the autonomous on-road vehicle **10**, to a certain location; autonomously release the cargo load **11** at the certain location (the transition from FIG. **3D** to FIG. **3C** to FIG. **3B** and to FIG. **3A**); self-drive the on-road autonomous vehicle **10** (FIG. **2A**), which is now free of the cargo load, to the certain parking location; upon arrival of the on-road autonomous vehicle **10** to said certain parking location: straddle autonomously, the on-road autonomous vehicle, over the cabin, thereby allowing the on-road autonomous vehicle to grab and lift autonomously the cabin; self-drive the on-road autonomous vehicle to a pick-up location, while the cabin is hanging underneath the on-road autonomous vehicle; and pick-up a passenger at the pick-up location (FIG. **5C**) into the cabin.

[0184] One embodiment is a system operative to transport both passengers in a towed cabin and

towed cargo loads, comprising: an autonomous on-road vehicle operative to tow loads such as towed cabins operative to contain passengers and such as towed cargo loads; a towed cabin operative to contain passengers, in which the towed cabin is currently located at a certain parking location; and a towed cargo load; wherein, the system is configured to: autonomously transport the towed cargo load, using the autonomous on-road vehicle, to a certain location; autonomously release the towed cargo load at the certain location; self-drive the on-road autonomous vehicle, which is now free of the towed cargo load, to the certain parking location; upon arrival of the on-road autonomous vehicle to said certain parking location: connect the towed cabin, thereby allowing the on-road autonomous vehicle to autonomously tow the towed cabin; self-drive the on-road autonomous vehicle to a pick-up location, while the towed cabin is hanging behind the on-road autonomous vehicle; and pick-up a passenger at the pick-up location into the cabin.

[0185] FIG. 13A illustrates one embodiment of an on-road autonomous vehicle **10** currently integrated with a first object **11-obj-1** having drawers **11-drawer** that are presently closed. The drawers **11-drawer** may contain packages to be delivered or other items to be delivered and/or stored. The on-road autonomous vehicle **10** is operative to self-integrate with the first object **11-obj-1**, thereby creating a first resultant vehicle **10+11-obj-1** that is the combination of both the autonomous vehicle **10** and the first object **11-obj-1** and that has a first specific purpose, in which such a first specific purpose may be the delivery of packages. For example, the first resultant vehicle **10+11-obj-1** may self drive to a logistics center, load the drawers **11-drawer** with packages, and then self-drive with the packages onboard to several delivery destinations, in which each of the drawers **11-drawer** gets open as the resultant vehicle **10+11-obj-1** arrives at the designated delivery destination, and the respective package is then picked up, perhaps by a certain person who ordered the respective package. Self integration of on-road autonomous vehicle **10** with the first object **11-obj-1** facilitates the first specific purpose of package delivery by: (i) combining the features of both the on-road autonomous vehicle **10** and the first object **11-obj-1** in creation of a combined set of features that facilitates self package delivery, and (ii) maintaining cooperation/communication between the on-road autonomous vehicle **10** and the first object **11-obj-1** during the different phases of package delivery. For example, self-driving the first resultant vehicle **10+11-obj-1** to the logistics center is based on sensors (e.g., **4a**, **4b**, **4c**, FIG. 1A) and self-driving directives embedded in the on-road autonomous vehicle **10**, but loading packages into the drawers **11-drawer** requires further cooperation and communication with the first object **11-obj-1**, for example when the sensors onboard the on-road autonomous vehicle **10** conclude that the first resultant vehicle **10+11-obj-1** is in the correct loading spot in the logistics center and consequently the on-road autonomous vehicle **10** instructs the first object **11-obj-1** to open all drawers **11-drawer** thereby allowing the loading of packages into the first object **11-obj-1**. When the drawers **11-drawer** are loaded with packages, the first resultant vehicle **10+11-obj-1** again uses sensors and directives embedded in the on-road autonomous vehicle **10** to self drive to several delivery locations. Upon arrival to a certain delivery location, the sensors **4a**, **4b**, **4c** onboard on-road autonomous vehicle **10** can be used again to identify a specific person to whom one of the packages is intended, and the on-road autonomous vehicle **10** can then order a particular one of the drawers **11-drawer** in the first object **11-obj-1** to open up and allow this person access to the respective package. As is evident from the above example, a synergy is created between the on-road autonomous vehicle **10** and the first object **11-obj-1**, in which such a synergy is the result of a deep integration between the on-road autonomous vehicle **10** and the first object **11-obj-1**, which now operate together as a synchronized and managed single resultant vehicle **10+11-obj-1**. It is noted that after said self integration of (i.e., deep integration between) the on-road autonomous vehicle **10** and the first object **11-obj-1**, the resultant vehicle **10+11-obj-1** has a resultant specific outer shape, for example by now including drawers **11-drawer** that are accessible to people standing alongside the resultant vehicle **10+11-obj-1**, in which such a resultant specific outer shape is critical for achieving the first specific purpose of package delivery. In one embodiment, an interface **5-int-1** is included in the on-road autonomous

vehicle **10**, in which maintaining the cooperation and communication between the on-road autonomous vehicle **10** and the first object **11-obj-1** is done via such interface. The interface **5-int-1** may support power transfer, such as electricity, from the on-road autonomous vehicle **10** to the first object **11-obj-1** and vice versa, and may further support data communication between the two, in which commands can be relayed through such interface and sensory data can be shared. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the first object **11-obj-1** is achieved via a straddling process, in which the on-road autonomous vehicle **10** straddles over the first object **11-obj-1**, lowers itself or a connector thereof **5-cnct**, grabs the first object **11-obj-1**, and lifts the first object **11-obj-1**, as described by the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, and in which the first object **11-obj-1** is represented by the label **11**.

[0186] FIG. 13B illustrates one embodiment of the on-road autonomous vehicle **10** still integrated with the first object **11-obj-1** and getting one of the drawers **11-drawer** opened, perhaps for loading or unloading a package into the drawer **11-drawer**.

[0187] FIG. 14A illustrates one embodiment of an on-road autonomous vehicle **10** currently integrated with a second object **11-obj-2** having a door **11-door** that is presently closed. The door **11-door** may allow passengers to get in and out of object **11-obj-2**, which may be a passenger cabin. The on-road autonomous vehicle **10** is operative to self-integrate with the second object **11-obj-2**, thereby creating a second resultant vehicle **10+11-obj-2** that is the combination of both the autonomous vehicle **10** and the second object **11-obj-2** and that has a second specific purpose, in which such a second specific purpose may be transporting passengers in conjunction with a taxi service. For example, the second resultant vehicle **10+11-obj-2** may self drive to a pick-up destination, in which the door **11-door** gets open as the second resultant vehicle **10+11-obj-2** arrives at the pick-up destination, and a passenger can then get into the passenger cabin **11-obj-2**. Self integration of on-road autonomous vehicle **10** with the second object **11-obj-2** facilitates the second specific purpose of transporting passengers by: (i) combining the features of both the on-road autonomous vehicle **10** and the second object **11-obj-2** in creation of a combined set of features that facilitates self transporting of passengers, and (ii) maintaining cooperation/communication between the on-road autonomous vehicle **10** and the second object **11-obj-2** during the different phases of transporting passengers. For example, self-driving the second resultant vehicle **10+11-obj-2** to the pick-up location is based on sensors (e.g., **4a**, **4b**, **4c**, FIG. 1A) and self-driving directives embedded in the on-road autonomous vehicle **10**, but letting a passenger get onboard the passenger cabin **11-obj-2** by opening the door **11-door** requires further cooperation and communication with the second object **11-obj-2**, for example when the sensors onboard the on-road autonomous vehicle **10** conclude that the second resultant vehicle **10+11-obj-2** is in the correct address and further recognizes the passenger to be picked-up, then consequently the on-road autonomous vehicle **10** instructs the second object **11-obj-2** to open the door **11-door** thereby allowing the recognized passenger in. As is evident from the above example, a different synergy is created between the on-road autonomous vehicle **10** and the second object **11-obj-2** (as compared to the synergy created with the first object **11-obj-1**), in which such a different synergy is the result of a deep integration between the on-road autonomous vehicle **10** and the second object **11-obj-2**, which now operate together as a differently synchronized and managed single second resultant vehicle **10+11-obj-2**. It is noted that after said self integration of the on-road autonomous vehicle **10** and the second object **11-obj-2**, the second resultant vehicle **10+11-obj-2** has a second resultant specific outer shape (as compared to the first resultant outer shape in conjunction with the first object **11-obj-1**), for example by now including a door **11-door** that is located at the right height allowing passengers in the resultant second vehicle **10+11-obj-2**, in which such a second resultant specific outer shape is critical for achieving the second specific purpose of transporting passengers. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the second object **11-obj-2** is achieved via a straddling process, in which the on-road autonomous vehicle **10** straddles over the second object **11-obj-2**, lowers itself or a connector thereof **5-cnct**, grabs the second object **11-obj-**

2, and lifts the second object **11-obj-2**, as described by the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, and in which the second object **11-obj-2** is represented by the label **11**. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the second object **11-obj-2** is done after de-integrating the on-road autonomous vehicle **10** with the first object **11-obj-1**, which may be achieved autonomously using a straddling-off process, in which the on-road autonomous vehicle **10** lowers the second object **11-obj-2**, and straddles off the second object **11-obj-2**, as can be visualized by “playing in reverse” the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, i.e., starting from FIG. 3D and ending with FIG. 3A.

[0188] FIG. 14B illustrates one embodiment of the on-road autonomous vehicle **10** still integrated with the second object **11-obj-2** and getting the door **11-door** opened, perhaps for letting passengers get into and out of the passenger cabin **11-obj-2**.

[0189] One embodiment is a system operative to autonomously alter functionality of an on-road vehicle. The system includes: an on-road vehicle **10** (FIG. 13A, FIG. 13B, FIG. 14A, FIG. 14B), operative to straddle over objects **11**; a first object **11-obj-1** (FIG. 13A, FIG. 13B) operative to facilitate a first function when integrated with the on-road vehicle **10**, in which the first object **11-obj-1** is currently integrated with the on-road vehicle **10** (as shown in FIG. 13A, FIG. 13B), thereby currently enabling the on-road vehicle **10** together with the first object **11-obj-1** to perform said first function; and a second object **11-obj-2** (FIG. 14A, FIG. 14B) operative to facilitate a second function when integrated with the on-road vehicle **10**, in which the second object **11-obj-2** is currently located at a certain location (e.g., at location **20-p** as shown in FIG. 2B, in which object **11-obj-2** is represented by the label **11**).

[0190] In one embodiment, as a response to a specific request received in the system, the system is configured to autonomously alter functionality of the on-road vehicle **10** from a first functionality associated with the first function into a different functionality associated with the second function, in which as a part of said autonomous alteration and said response, the on-road vehicle is configured to: release autonomously the first object **11-obj-1**; self drive from a current location of the on-road vehicle **10** to said certain location **20-p** of the second object **11-obj-2**; upon arrival to said certain location **20-p**: straddle autonomously over the second object **11-obj-2** (e.g., as shown in FIG. 3A, FIG. 3B, in which object **11-obj-2** is represented by the label **11**), thereby allowing the on-road vehicle **10** to grab and lift autonomously the second object **11-obj-2** above ground (e.g., as shown in FIG. 3C, FIG. 3D), thereby integrating autonomously the second object **11-obj-2** with the on-road vehicle **10**; and perform said second function in conjunction with the second object **11-obj-2** now integrated with the on-road **10** vehicle (as shown in FIG. 14A, 14B).

[0191] In one embodiment, the first object **11-obj-1** is essentially a first type of container having a surface with a first type of interface **11-drawer** (FIG. 13A, FIG. 13B), in which the first type of interface is operative to facilitate a certain first way of interfacing with people; and the second object **11-obj-2** is essentially a second type of container having a surface with a second type of interface **11-door** (FIG. 14A, FIG. 14B), in which the second type of interface is operative to facilitate a certain second way of interfacing with people.

[0192] In one embodiment, the first type of container **11-obj-1** is a container operative to contain packages in drawers **11-drawer**; the first function is autonomous package delivery; the first type of interface is associated with at least one of the drawers **11-drawer** getting opened (FIG. 13A, FIG. 13B); the first way of interfacing with people comprises people accessing the drawers **11-drawer** and collecting a package delivered by the on-road vehicle **10**; and the first function of autonomous package delivery is facilitated by said integration of the first object **11-obj-1** with the on-road vehicle **10**, in which said integration enables the system to both: (i) facilitate said delivery by driving autonomously the on-road vehicle **10** with packages onboard, and (ii) facilitate said collection of the packages by people accessing the drawers **11-drawer**.

[0193] In one embodiment, the second type of container **11-obj-2** is a container operative to accommodate passengers; the second function is an autonomous taxi service; the second type of

interface is associated with a door **11-door** (FIG. 14A, FIG. 14B) operative to allow the passengers getting into and out-of the second type of container **11-obj-2**; the second way of interfacing with people comprises people opening and closing the door **11-door**; and the second function of autonomous taxi service is facilitated by said integration autonomously of the second object **11-obj-2** with the on-road vehicle **10**, in which said integration autonomously enables the system to both: (i) facilitate said taxi service by autonomously transporting the passengers by the on-road vehicle **10** and in conjunction with the second object **11-obj-2**, and (ii) further facilitate said taxi service by said allowing the passengers to get into and out-of the second type of container **11-obj-2** using the door **11-door**.

[0194] In one embodiment, said lifting autonomously of the second object **11-obj-2** above ground is done so as to position the door **11-door** at a certain height above ground that is operative to allow said passengers getting into and out-of the second type of container **11-obj-2**, in which said certain height is between 20 (twenty) centimeters and 70 (seventy) centimeters above ground.

[0195] In one embodiment, the second object **11-obj-2** is selected from a group consisting of: (i) a container operative to contain packages in drawers **11-drawer**, in which the second function is autonomous package delivery, (ii) a container operative to accommodate passengers **30** (FIG. 5B), in which the second function is a taxi service, (iii) a container operative to contain a load, in which the second function is transporting loads, (iv) a power source such as battery (e.g., **11-func**, FIG. 7B), a fuel cell, and a generator, in which the second function is charging the on-road vehicle **10** while on the move, (v) a mobile vending machine, in which the second function is selling goods at different locations, (vi) a tank, in which the second function is transporting substances such as liquids and compressed gas, (vii) transportable electronic communication equipment such as a radio access network (RAN), in which the second function is providing electronic communication services from different locations.

[0196] In one embodiment, the on-road vehicle **10** comprises: an upper horizontal structure **3a**, **3b**, **3c** (FIG. 1A) elevated above ground by vertical structures **2a**, **2a'**, **2b**, **2b'**, **2c**, **2c'**, **2d**, **2d'** (FIG. 1A) mounted on at least four wheels **1a**, **1b**, **1c**, **1d** (FIG. 1A) touching ground **9-ground**, so as to create a certain clearance **9-clr** above ground for at least a first connector **5-cnct** associated with the upper horizontal structure **3b** and attached thereunder; a control sub-system **4**, **6**, **7**, **8** comprising a processing unit **8** (FIG. 1D) and a plurality of sensors **4a**, **4b**, **4c** (FIG. 1A) and actuators **6**, **7** (FIG. 1D), in which the control sub-system is configured to generate, in real-time, a three-dimensional representation of surrounding environment using data collected by the plurality of sensors; and at least a first linear actuator **2'+2** (**2'** moving up and down relative to **2**, i.e., **2a'** moving relative to **2a**, **2b'** moving relative to **2b**, **2c'** moving relative to **2c**, and **2d'** moving relative to **2d**, FIG. 1A) configured to control and set said certain clearance **9-clr** of the first connector **5-cnct**, by causing the first connector, or the entire upper horizontal structure **3a**, **3b**, **3c** including the first connector, to move up or down relative to ground **9-ground**. In one embodiment, the control sub-system **4**, **6**, **7**, **8** is further configured to use said three-dimensional representation, said actuators **6**, **7**, and said processing unit **8** in conjunction with a set of public-road self-driving directives, to: (i) facilitate said self-driving (FIG. 2A, FIG. 2B) of the on-road vehicle **10**, over public roads **20a** (FIG. 2A), **20b** (FIG. 2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f** (FIG. 2A, FIG. 2B, FIG. 2C), to said certain location **20-p**, (ii) position the on-road autonomous vehicle **10** in front of the second object **11-obj-2** (**11** in FIG. 2C, FIG. 3A), and (iii) facilitate said straddling of the on-road vehicle **10** over the second object **11-obj-2** (**11** in FIG. 3B), such that said first connector **5-cnct** is brought to a predetermined position **5-pos** over the second object **11-obj-2** (**11** in FIG. 3B); and the control sub-system **4**, **6**, **7**, **8** is further configured to use the first linear actuator **2'+2** to: (i) facilitate said grabbing by lowering the first connector **5-cnct** into mechanical contact with the second object **11-obj-2** (**11** in FIG. 3C) thereby allowing the connector to connect to or otherwise grab the second object, and (ii) facilitate said lifting by lifting (FIG. 3D) the second object **11-obj-2** above ground into a position operative to self-transport the second object.

[0197] In one embodiment, said first linear actuator 2'+2 is a distributed linear actuator comprising several sub-actuators 2a, 2a', 2b, 2b', 2c, 2c', 2d, 2d' (FIG. 1A), in which each sub-actuator is associated with one of the wheels 1a, 1b, 1c, 1d (FIG. 1A), such that the entire upper horizontal structure 3a, 3b, 3c is operative to move up and down relative to the wheels 1a, 1b, 1c, 1d and ground 9-ground, in which the linear actuators 2'+2 are also operative to act as springs/mechanical dampers for the wheels 1a, 1b, 1c, 1d relative to the upper horizontal structure 3a, 3b, 3c.

[0198] In one embodiment, said first linear actuator 2'+2 is embedded in the first connector 5-cnct, thereby causing only the connector to move up and down relative to ground 9-ground, and such that the upper horizontal structure 3a, 3b, 3c remains in place.

[0199] In one embodiment, said integration of the first object 11-obj-1 with the on-road vehicle 10 was previously achieved by the on-road vehicle by performing a previous autonomous maneuver as a response to a particular request received in the system, in which as part of the previous autonomous maneuver, the on-road vehicle 10 is configured to: self drive from a previous location of the on-road vehicle to a specific location at which the first object 11-obj-1 is located; upon arrival to said specific location: straddle autonomously over the first object 11-obj-1, thereby allowing the on-road vehicle 10 to grab and lift autonomously the first object above ground, thereby facilitating said integration of the first object 11-obj-1 with the on-road vehicle 10.

[0200] In one embodiment, as a part of said releasing autonomously of the first object 11-obj-1, the on-road vehicle 10 is configured to: (i) lower the first object, (ii) un-grab the first object and (iii) straddle off the first object.

[0201] In one embodiment, said autonomously altering of functionality is facilitated by a combination of different autonomous functions working in synchronization, in which said combination of different autonomous functions comprises: (i) said self-driving, thereby allowing the on-road vehicle 10 to autonomously access the second object 11-object-2, (ii) said straddling autonomously, thereby allowing the on-road vehicle 10 to self-align with the second object 11-object-2, and (iii) said grabbing and lifting autonomously of the second object 11-object-2, thereby allowing the on-road vehicle 10 to self-integrate with the second object, in which said autonomously altering of functionality is further facilitated by an ability of the on-road vehicle 10 to straddle over the respective objects 11-object-1, 11-object-2, and in which said autonomously altering of functionality comprises at least said alteration of functionality taking place without relying on external support such as support from people for driving, support from people for lifting and displacing/aligning loads, and support from external mechanical devices for lifting and displacing/aligning loads; and said autonomously altering of functionality is further facilitated by the on-road vehicle eclectically interfacing 5-int-1 with the respective objects, in which: said electrically interfacing is done in conjunction with said grabbing of the respective object; said electrically interfacing comprises at least one of: (i) supplying electrical power to the respective object, (ii) supplying communication services to the respective object, and (iii) supplying sensory information to the respective object, thereby allowing the respective object to better interact with people in conjunction with accomplishing the respective functionality; and said electrically interfacing 5-int-1 is a part of said integration.

[0202] In one embodiment, said autonomously altering of functionality comprises switching/changing/modifying autonomy mode by the on-road vehicle, from a first autonomy mode into a second autonomy mode, in which the first autonomy mode is operative to support a first automatic on-road behavior that facilitates the first function, and the second autonomy mode is operative to support a second and different automatic on-road behavior that facilitates the second function, in which said switching/changing of the autonomy mode is a part of said integration. In one embodiment, the first object 11-obj-1 is a container operative to contain packages in drawers 11-drawer; the first function is autonomous package delivery; and the first autonomy mode is a mode that automatically facilitates arriving with the packages to a vicinity of people and enabling the people to access the drawers 11-drawer and collect a package delivered by the on-road vehicle

10. In one embodiment, the second object **11-obj-2** is a container operative to accommodate passengers; the second function is an autonomous taxi service; and the second autonomy mode is a mode that facilitates said taxi service by autonomously transporting the passengers by the on-road vehicle **10**, and autonomously allowing the passengers to get into and out-of the second type of container **11-obj-2** using a door **11-door**.

[0203] FIG. **15** illustrates one embodiment of a method for autonomously altering functionality of an on-road vehicle. The method includes: In step **1101**, performing a first function by an on-road vehicle **10**, in which the first function is performed by the on-road vehicle in conjunction with a first object **11-obj-1** that is currently integrated with the on-road-vehicle and that is operative to facilitate said first function. In step **1102**, receiving, in conjunction with the on-road vehicle **10**, a request associated with altering functionality of the on-road vehicle from a first functionality associated with the first function into a different functionality associated with a second function. In step **1103**, releasing autonomously the first object **11-obj-1** by the on-road vehicle **10** as a response to said request. In step **1104**, self driving by the on-road vehicle **10**, as a further response to said request, from a current location of the on-road vehicle to a certain location at which a second object **11-obj-2** is located, in which the second object is associated with said different functionality. In step **1105**, upon arrival to said certain location: (i) straddling autonomously, by the on-road vehicle **10**, over the second object **11-obj-2**, (ii) grabbing and lifting, autonomously, the second object **11-obj-2** above ground by the on-road vehicle **10**, and thereby autonomously integrating the second object **11-obj-2** with the on-road vehicle **10**. In step **1106**, performing said second function by the on-road vehicle in conjunction with the second object **11-obj-2** now integrated with the on-road vehicle **10**.

[0204] In one embodiment, the method further comprises: receiving, in conjunction with the on-road vehicle **10**, and prior to said performing of the first function, a prior request associated with altering functionality of the on-road vehicle into the first functionality associated with the first object **11-obj-1**; self driving by the on-road vehicle **10**, as a response to said prior request, from a previous location of the on-road vehicle to a particular location at which the first object **11-obj-1** is located; and upon arrival to said particular location: (i) straddling autonomously, by the on-road vehicle **10**, over the first object **11-obj-1**, (ii) grabbing and lifting, autonomously, the first object **11-obj-1** above ground by the on-road vehicle **10**, thereby facilitating said integration of the first object **11-obj-1** with the on-road vehicle **10** and said performing of the first function by the on-road vehicle **10**.

[0205] In one embodiment, said releasing autonomously of the first object **11-obj-1** by the on-road vehicle **10** comprises: lowering the first object **11-obj-1** by the on-road vehicle **10**; un-grabbing the first object **11-obj-1** by the on-road vehicle **10**; and straddling off the first object **11-obj-1** by the on-road vehicle **10**.

[0206] One embodiment is a system operative to respond to a dynamic demand for various functions by autonomously altering shape and functionality of on-road vehicles. The system includes: a fleet of on-road vehicles comprising a plurality of on-road vehicles such as vehicle **10**, in which each of the on-road vehicles **10** is configured to autonomously-upon-demand pick-up and integrate-with various objects **11-obj-1**, **11-obj-2**; and a pool of objects **11-obj-1**, **11-obj-2** comprising said various objects, in which each of the objects is associated with a respective functionality, thereby collectively supporting a variety of functionalities, and in which there are more objects **11-obj-1**, **11-obj-2** in the pool than on-road vehicles **10** in the system.

[0207] In one embodiment, the system is configured to: determine current demands for various functionalities; determine, based on said current demands, a new assignment of functionalities for at least some of the on-road vehicles **10**, in which said new assignment is expected to allow the fleet of on-road vehicles to better respond to the demands; and per each of the on-road vehicles **10** for which a new assignment of functionality was determined, the on road vehicle **10** is configured to: (i) release one of the objects **11-obj-1** that is currently integrated therewith, (ii) self drive from a

current location of the on-road vehicle **10** to a location of parking of another one of the objects **11-obj-2** that is associated with the respective functionality newly assigned, and (iii) upon arrival to the location of parking: straddle autonomously over the another object **11-obj-2**, thereby allowing the on-road vehicle **10** to grab and lift autonomously said another object **11-obj-2** above ground, and thereby integrating autonomously the another object **11-obj-2** with the on-road vehicle **10**, thus embedding in the on-road vehicle **10** the respective functionality newly assigned.

[0208] In one embodiment, per each of the on-road vehicles **10** for which a new assignment of functionality was determined: said integrating autonomously of the respective another object **11-obj-2** with the on-road vehicle **10** results in an alteration of an outer shape of the on-road vehicle **10** from a previous outer shape associated with the respective previously integrated object **11-obj-1** into a new outer shape associated with the respective newly integrated object **11-obj-2**.

[0209] In one embodiment, said alteration of the outer shape facilitates said new functionality assigned.

[0210] In one embodiment, said previous outer shape is associated with a first outer interface (e.g., **11-drawer**) in the previously integrated object **11-obj-1**, in which said first outer interface is associated with a first way of interacting with people; and said new outer shape is associated with a second outer interface (e.g., **11-door**) in the newly integrated object **11-obj-2**, in which said second outer interface is associated with a second way of interacting with people.

[0211] FIG. **16** illustrates several embodiments of on-road autonomous vehicles having several different sizes respectively. A large on-road autonomous vehicle **10-large** is operative to straddle over and pick up large-sized containers/loads **11-large**, in which large-sized containers are containers having at least one dimension (e.g., length) that is longer than 2 (two) meters, and in which such containers are operative to be used in conjunction with various functions, such as transporting passengers and transporting/delivering packages. A medium on-road autonomous vehicle **10-medium** is operative to straddle over and pick up medium-sized containers/loads **11-medium**, in which medium-sized containers are containers having at least one dimension between 1 (one) meter and 2 (two) meters, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering groceries and packages. A small on-road autonomous vehicle **10-small** is operative to straddle over and pick up small-sized containers/loads **11-small**, in which small-sized containers are containers having at least one dimension between 50 (fifty) centimeters and 1 (one) meter, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering packages and delivering food. In one embodiment, the small load **11-small** is, by itself, a package being delivered by the small on-road autonomous vehicle **10-small**. A tiny on-road autonomous vehicle **10-tiny** is operative to straddle over and pick up tiny containers/loads **11-tiny**, in which tiny containers are containers having at least one dimension between 20 (twenty) centimeters and 50 (fifty) centimeters, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering packages, documents, and medicine. In one embodiment, the tiny load **11-tiny** is, by itself, a package being delivered by the small on-road autonomous vehicle **10-small**.

[0212] FIG. **17A** illustrates one embodiment of a wheel robot **10** carrying a disposable package **11-crb** using straps **92-strap**.

[0213] FIG. **17B** illustrates one embodiment of a wheel robot **10** carrying a disposable package **11-crb** by inserting grabbing elements **5-cnct** into handle holes **92-ph** embedded in the package.

[0214] FIG. **17C** illustrates one embodiment of a wheel robot **10** carrying a disposable package **11-crb** using side arms **5-side-arm** operative to move sideways to release the package.

[0215] FIG. **17D** illustrates one embodiment of a wheel robot **10** carrying a disposable package **11-bag** comprising a bag.

[0216] One embodiment is a system operative to autonomously deliver consumer items in disposable packages. The system includes: a wheel robot **10** (FIG. **17A**) operative to straddle over

and off objects; at least one type of disposable package **11** (FIG. 17A) comprising a respective type of disposable handle **92** (FIG. 17A); and a grabbing mechanism 5-cnct (FIG. 17A) embedded in the wheel robot **10**.

[0217] In one embodiment, the wheel robot **10** is configured to carry a consumer item **93-ci** (FIG. 17A) in said disposable package **11** from a pick-up location to a drop-off location, in which the grabbing mechanism 5-cnct is configured to grab the disposable package **11** by the respective disposable handle **92** and under the wheel robot during said carriage. Upon arrival of the wheel robot **10** to the drop-off location, the grabbing mechanism 5-cnct is configured to un-grab the disposable package **92**, thereby releasing the disposable package on ground under the wheel robot (e.g., on a sidewalk, on a road, on grass, or on a flat object placed on ground); and the wheel robot **10** is further configured to straddle off the disposable package **92** and the consumer item **93-ci**, thereby leaving the consumer item in the disposable package on the ground for a consumer to pick up.

[0218] In one embodiment, the disposable package **11** is of a cardboard packaging box type **11-crb** (FIG. 17A, FIG. 17B, FIG. 17C). In one embodiment, the respective disposable handle type **92** comprises at least one of: (i) at least one strap **92-strap** (FIG. 17A) attached to the top of the cardboard packaging box **11-crb**, (ii) punch-holes **92-ph** (FIG. 17B, FIG. 17C) in opposite two sides of the cardboard packaging box **11-crb**, (iii) a cardboard handle element at the top of the cardboard packaging box, and (iv) a rope and/or a string and/or a strap around and/or through the cardboard packaging box.

[0219] In one embodiment, the disposable package **11** is of a plastic bag type **11-bag** (FIG. 17D). In one embodiment, the respective disposable handle type **92** is an integral part **92-in** (FIG. 17D) of the plastic bag **11-bag**. In one embodiment, the plastic bag **11-bag** is made of a material comprising at least one of: (i) high-density polyethylene (HDPE), (ii) low-density polyethylene (LDPE), and (iii) linear low-density polyethylene (LLDPE). In one embodiment, the plastic bag **11-bag** is made of a plastic net, and the respective disposable handle type **92** comprises a part of the plastic net. In one embodiment, the plastic bag **11-bag** is made of polyvinyl chloride (PVC) clear semi-rigid plastic structure.

[0220] In one embodiment, the disposable package **11** is of a paper bag type, in which the respective disposable handle type **92** comprises at least one of: (i) punch-holes in the paper bag, and (ii) at least one strap and/or a string attached to the paper bag.

[0221] In one embodiment, the disposable package **11** is of a cloth and/or canvas bag type, in which the respective disposable handle type comprises at least one strap and/or string attached to the top of the cloth and/or canvas bag.

[0222] In one embodiment, the wheel robot **10** is small in dimensions relative to an average on-road vehicle, measuring less than 2.5 (two point five) meters in length, less than 1.5 (one point five) meters in height, and weighing less than 700 (seven hundred) kilograms, thereby allowing the wheel robot access to sidewalks. In one embodiment, the drop-off location is situated on a sidewalk and in proximity to an address associated with the consumer; and the wheel robot **10** is configured to travel on the sidewalk while avoiding pedestrians. In one embodiment, the disposable package **11** comprises a cardboard packaging box **11-crb**, and prior to said releasing of the disposable package with the consumer item inside, the wheel robot is configured to detect that the drop-off location is at least dry, thereby assuring survivability of the cardboard packaging box and the consumer item inside. In one embodiment the cardboard packaging box **11-crb** comprises a disposable extension attached to a bottom side of the cardboard packaging, thereby assuring survivability of the cardboard packaging when the drop-off location is wet. In one embodiment, said extension comprises cardboard structures such as cardboard legs.

[0223] In one embodiment, the wheel robot **10** is small in dimensions relative to an average human being, measuring less than 1.5 (one point five) meters in length, less than 1 (one) meter in height, and weighing less than 200 (two hundred) kilograms; the disposable package **11** is small in

dimensions relative to an average human being, has at least one dimension of less than 1 (one) meter, and is therefore carriable by the average human being; and the disposable package **11** comprises at least one of: (i) a plastic bag **11-bag**, (ii) a paper bag, (iii) a cardboard packaging box **11-crb**, and (iv) a polystyrene foam structure. In one embodiment, the wheel robot **10** is configured to move at speeds of less than 2 (two) meters per second, thereby increasing safety in conjunction with said sidewalk travel and pedestrian avoidance. In one embodiment, the disposable handle **92** is dual purposed, in which a first purpose is associated with said grabbing mechanism **5-cnct** and the second purpose is associated with the consumer picking up the disposable package **11** with the consumer item **93-ci** inside.

[0224] In one embodiment, prior to said carriage from the pick-up location to the drop-off location: the wheel robot **10** is further configured to straddle over the disposable package **11** containing the consumer item **93-ci** at the pick-up location; and [0225] the grabbing mechanism **5-cnct** is further configured to grab the disposable package **11** by the respective disposable handle **92**.

[0226] FIG. **18** illustrates one embodiment of a method for autonomously delivering consumer items in disposable packages. The method includes: In step **1111**, receiving an order to deliver a consumer item **93-ci** (FIG. **17A**, FIG. **17D**) to a consumer. In step **1112**, grabbing, by a grabbing mechanism **5-cnct** (FIG. **17A**) in a wheel robot **10** (FIG. **17A**), a disposable package **11** (FIG. **17**) containing the consumer item **93-ci** at a pick-up location. In step **1113**, carrying, by a wheel robot **10**, the consumer item **93-ci** in the disposable package **11** from the pick-up location to a drop-off location, in which the grabbing mechanism **5-cnct** is configured to grab the disposable package during said carriage and under the wheel robot. In step **1114**, upon arrival of the wheel robot **10** to the drop-off location, un-grabbing the disposable package **11**, thereby releasing the disposable package on ground under the wheel robot. In step **1115**, straddling, by the wheel robot **10**, off the disposable package **11** and the consumer item **93-ci**, thereby leaving the consumer item in the disposable package on the ground for a consumer to pick up.

[0227] In one embodiment, said grabbing is done so as to place a bottom of the disposable package **11** at a height of no more than 1 (one) meter above ground, and no less than 10 (ten) centimeters above ground during said carriage; and said releasing results in said disposable package **11** with the consumer item inside **93-ci** falling a short distance of between 10 centimeters and one meter to the ground, in which a survival of the consumer item when the disposable package hits the ground is facilitates by at least one of: (i) said distance being short, and (ii) said disposable package comprising a soft and/or collapsible cushioning material **94-cm** (FIG. **17D**) that is an integral part of the disposable package **11** and/or that is attached to the bottom side of the disposable package.

[0228] In one embodiment, the method further comprises prior to said grabbing: straddling, by the wheel robot **10**, over the disposable package **11** containing the consumer item **93-ci** at the pick-up location; lowering the grabbing mechanism **5-cnct** to a position to perform said grabbing; and lifting above ground, by the grabbing mechanism **5-cnct**, the disposable package **11** together with the consumer item **93-ci**; in which said releasing comprises the grabbing mechanism **5-cnct** lowering the disposable package **11** together with the consumer item **93-ci** to the ground prior to said un-grabbing.

[0229] In one embodiment, the consumer item **93-ci** comprises at least one of: (i) groceries, (ii) a home appliance, (iii) clothing, (iv) a consumer electronic device, (v) books and/or other paper products, (vi) a pharmaceutical product, (vii) a toy, (viii) a takeout item, (ix) a parcel, (x) a kitchenware item, (xi) a furniture item, (xii) a care product, (xiii) a beauty product, (xiv) a toiletry product, and (xv) a tool item.

[0230] FIG. **19A** illustrates one embodiment of an on-road autonomous vehicle **10** carrying a first package container **11m-4**. The on-road vehicle **10** is operative to straddle-over and grab the package container using a grabbing mechanism **5-cnct-8** in accordance with some embodiments. A second grabbing mechanism **5-cnct-9** is also shown.

[0231] FIG. **19B** illustrates one embodiment of the on-road autonomous vehicle **10** carrying the

first package container **11m-4** and a second package container **11m-5** at the same time and in tandem. The on-road vehicle **10** is operative to straddle-over and grab the second package container **11m-5** using a grabbing mechanism **5-cnct-9** in accordance with some embodiments. The on-road vehicle **10** may straddle-over and grab the second package container **11m-5** via **5-cnct-9** while the first package container **11m-4** is already attached via **5-cnct-8**. The on-road vehicle **10** may straddle-over and grab package container **11m-4** first and then straddle-over and grab package container **11m-5**, or grab package container **11m-5** first and then straddle-over and grab package container **11m-4**. The on-road vehicle **10** is operative to release and straddle-off the second package container **11m-5** in accordance with some embodiments. The on-road vehicle **10** may release and straddle-off the second package container **11m-5** while the first package container **11m-4** is still attached via **5-cnct-8**. The on-road vehicle **10** may release and straddle-off package container **11m-4** first and then release and straddle-off package container **11m-5**, or release and straddle-off container **11m-5** first and then release and straddle-off package container **11m-4**.

[0232] In one embodiment, the on-road vehicle **10** may transport only one of the package containers in some occasions, as shown in FIG. **19A**, and may transport two or more of the package containers in other occasions as shown in FIG. **19B**, according to a current need and in conjunction with some embodiments.

[0233] It is noted that on-road vehicle **10** is depicted as carrying two of the packages **11m-4**, **11m5**, but on-road vehicle **10** may carry more packages in tandem using additional grabbing mechanisms **5-cnct** (not shown), or it may carry more packages side-by-side rather than in tandem (not shown).

[0234] FIG. **19C** illustrates one embodiment of the first package container **11m-4** comprising a plurality of lockers **11L-41**, **11L-42** containing a respective plurality of packages **11pkg-41**, **11pkg-42**, in which each of the packages in the lockers **11L-41**, **11L-42** is accessible via a respective locker door **11dr-41**, **11dr-42**. The package container **11m-4** includes a grabbing point **5-grb-4** via which the on-road vehicle **10** is able to grab and lift the container, using **5-cnct-8** for example, into a transport position as illustrated in FIG. **19A**.

[0235] FIG. **19D** illustrates one embodiment of the second package container **11m-5** comprising a plurality of lockers **11L-51**, **11L-52** containing a respective plurality of packages **11pkg-51**, **11pkg-52**, in which each of the packages in the lockers **11L-51**, **11L-52** is accessible via a respective locker door **11dr-51**, **11dr-52**. The package container **11m-5** includes a grabbing point **5-grb-5** via which the on-road vehicle **10** is able to grab and lift the container, using **5-cnct-9** for example, into a transport position as illustrated in FIG. **19B**.

[0236] In one embodiment, each of the lockers **11L-41**, **11L-42**, **11L-51**, **11L-52** may include a locking mechanism, e.g., the locking mechanism **11CLK** of locker **11L-52** is shown. The locking mechanisms may be individually commanded externally to allow unlocking of a respective one of the doors **11dr-41**, **11dr-42**, **11dr-51**, **11dr-52** in conjunction with a respective customer approaching the container to pick-up a respective one of the packages **11pkg-41**, **11pkg-42**, **11pkg-51**, **11pkg-52**.

[0237] FIG. **19E** illustrates one embodiment of the on-road autonomous vehicle **10** and package containers **11m** being in communicative contact **10-comm**, e.g., wireless contact, with a server **8-server** that may be external to vehicle **10**, e.g., located in a control center, or embedded in the vehicle, e.g., part of processing unit **8** (FIG. **1D**). The package containers **11m** may include a locally embedded processor **8-embd** operative to receive instruction and send updates from/to server **8-server**. Any operation and/or action executed in the computers **8**, **8-server**, **8-embd** may be facilitated and governed by a non-transitory and/or transitory machine readable code stored locally in conjunction with the respective computer using memory components such as flash memory, random-access-memory (RAM), and hard-disk-drives (HDD).

[0238] FIG. **19F** illustrates one embodiment of an example of a route **8-leg-1**, **8-leg-2**, **8-leg-3**, **8-leg-4**, **8-leg-5**, **8-leg-6**, **8-leg-7**, **8-leg-8** taken by the on-road autonomous vehicle **10** in conjunction with moving and placing the package containers **11m-4**, **11m-5** in different delivery locations **7-loc**,

as described in the following paragraphs:

[0239] According to one example, on-road vehicle **10** may first grab both of containers **11m-4**, **11m-5** from a location of origin 7-loc-0 associated, perhaps, with a warehouse and/or a store, in which the containers **11m-4**, **11m-5** are pre-loaded with the packages **11pkg-41**, **11pkg-42**, **11pkg-51**, **11pkg-52** to be delivered en-route 8-leg. On-road vehicle **10** then transports the containers **11m-4**, **11m-5**, together, to a first location 7-loc-1, perhaps a parking location near a house in which a first customer resides. On-road vehicle **10** then releases the first package container **11m-4** at location 7-loc-1, and server **8-server** receives **10-comm**, either from container **11m-4** or vehicle **10**, an indication that package **11pkg-41** is at location 7-loc-1, following which the server **8-server** may relate this information to the first customer, perhaps via his smartphone. However, vehicle **10** does not wait for the first customer to pick-up package **11pkg-41**, but instead continues on its journey 8-leg-2 toward location 7-loc-3 with the second package container **11m-5** still onboard, while leaving container **11m-4** at location 7-loc-1 together with package **11pkg-41** (and **11pkg-42** as well).

[0240] Upon arrival of vehicle **10** to location 7-loc-3, it releases the second package container **11m-5**, and server **8-server** receives an indication that package **11pkg-51** is at location 7-loc-3, following which the server **8-server** may relate this information to another customer associated with location 7-loc-3.

[0241] Then, upon an indication received in server **8-server**, that the first customer has picked up package **11pkg-41** from locker **11L-41** of container **11m-4** at location 7-loc-1, the vehicle **10** returns 8-leg-3 to location 7-loc-1, without any container onboard, to grab container **11m-4** and transport it 8-leg-4 to location 7-loc-2. Upon arrival of container **11m-4** to location 7-loc-2, server **8-server** receives an indication that package **11pkg-42** is at location 7-loc-2, following which the server **8-server** may relate this information to yet another customer associated with location 7-loc-2.

[0242] Then, upon an indication received in server **8-server**, that package **11pkg-51** has been picked up by the respective customer from locker **11L-51** of container **11m-5** at location 7-loc-3, the vehicle **10** returns 8-leg-5 to location 7-loc-3, without any container onboard, to grab container **11m-5** and transport it 8-leg-6 to location 7-loc-4. Upon arrival of container **11m-5** to location 7-loc-4, server **8-server** receives an indication that package **11pkg-52** is at location 7-loc-4, following which the server **8-server** may relate this information to yet another customer associated with location 7-loc-4.

[0243] Then, upon an indication received in server **8-server**, that packages **11pkg-52** and **11pkg-42** have been picked up by the respective customers from lockers **11L-42** of container **11m-4** at location 7-loc-2 and **11L-52** of container **11m-5** at location 7-loc-4, the vehicle **10** grabs container **11m-5** at location 7-loc-4, returns 8-leg-7 to location 7-loc-2, with container **11m-5** onboard, to grab container **11m-4** as well, and transport both containers 8-leg-8 back to the origin 7-loc-0. Upon arrival of containers **11m-4**, **11m-5** to the origin 7-loc-0, the containers may be loaded again with additional packages to be delivered in conjunction with a next delivery round.

[0244] One embodiment is a system operative to autonomously deliver packages using a plurality of package containers, comprising: at least a first **11m-4** (FIG. 19C) and a second **11m-5** (FIG. 19D) package containers, in which each of the package containers comprises a plurality of lockers **11L** (FIG. 19C, FIG. 19D) each comprising a respective door **11dr** (FIG. 19C, FIG. 19D) operative to be locked and unlocked independently of the other doors in the respective package container, and in which each of at least some of the lockers contains a respective package **11pkg** (FIG. 19C, FIG. 19D) to be delivered; and an on-road vehicle **10** (FIG. 19A) operative to straddle over the package containers **11m-4**, **11m-5**.

[0245] In one embodiment, upon reception of an indication pertaining to a delivery status associated with a first package **11pkg-41** in one of the lockers **11L-41** of the first container **11m-4** parked at a first location 7-loc-1 (FIG. 19F), the on-road vehicle **10** is configured to: straddle-over and grab the first container **11m-4** from the first location 7-loc-1, transport 8-leg-4 (FIG. 19F) the first container **11m-4** to a second location 7-loc-2, and straddle-off the first container to be left

parked at the second location; then upon reception of an indication pertaining to a delivery status associated with a second package **11pkg-51** in one of the lockers **11L-51** of the second container **11m-5** parked at a third location **7-loc-3**, the on-road vehicle **10** is configured to: straddle-over and grab the second container **11m-5** from the third location **7-loc-3**, transport **8-leg-6** the second container **11m-5** to a fourth location **7-loc-4**, and straddle-off the second container to be left parked at the fourth location; in which the system is configured to: in conjunction with arrival of the first container **11m-4** at the second location **7-loc-2**, send an indication that a third package **11pkg-42** is now and/or will be in place to be picked-up from another one of the lockers **11L-42** of the first container **11m-4**; and in conjunction with arrival of the second container **11m-5** at the fourth location **7-loc-4**, send an indication that a fourth package **11pkg-52** is now and/or will be in place to be picked-up from another one of the lockers **11L-52** of the second container **11m-5**.

[0246] In one embodiment, said indication pertaining to the delivery status associated with the first package **11pkg-41** is an indication that the first package was picked-up out of one of the lockers **11L-41** of the first container **11m-4** parked at the first location **7-loc-1**.

[0247] In one embodiment, said indication pertaining to the delivery status associated with the first package **11pkg-41** is an indication that the first package has been failed to be picked-up out of one of the lockers **11L-41** of the first container **11m-4** parked at the first location **7-loc-1**.

[0248] In one embodiment, each of the lockers **11L** is integrated in one of the sides of the respective package container **11m** so as to allow direct access to a person picking-up the respective package **11pkg**.

[0249] In one embodiment, each of said doors **11dr** constitutes a part of said side of the respective package container **11m**.

[0250] In one embodiment, at least one of the doors, e.g., **11dr-52**, is operative to be opened via a respective hinge in one of the sides of the door.

[0251] In one embodiment, at least one of the doors **11dr** is a part of drawer containing the respective package **11pkg**, e.g., similar to **11-drawer** in FIG. **13B**.

[0252] In one embodiment, each of the lockers **11L** is operative to be unlocked using an electronic lock mechanism **11LCK** (FIG. **19D**) embedded in the locker.

[0253] In one embodiment, upon reception of a respective unlock indication, the system is configured to electronically unlock the electronic lock mechanism **11LCK** of a certain one of the lockers **11L** in order to allow a respective authorized person to pick-up the respective package **11pkg**.

[0254] In one embodiment, each of the package containers **11m** has a weight of at least 50 (fifty) kilograms, so as to make it difficult for a single person to manually carry the package container, thereby preventing theft.

[0255] In one embodiment, each of the package containers **11m** has a weight of at least 100 (one hundred) kilograms, so as to make it difficult for a non-straddling on-road vehicle to transport the package container away from the respective parking location, thereby preventing theft.

[0256] In one embodiment, prior to said transports **8-leg-4**, **8-leg-6**, the on-road vehicle **10** is further configured to: straddle-over and grab the first package container **11m-4** parked at a warehouse location **7-loc-0**, as illustrated in FIG. **19A**; straddle-over and grab the second package container **11m-5** parked at the same warehouse location **7-loc-0**, thereby grabbing both the first and second package containers **11m-4**, **11m-5** at the same time and in tandem, as illustrated in FIG. **19B**; carry **8-leg-1** (FIG. **19F**) the first and second package containers **11m-4**, **11m-5** together to the first location **7-loc-1** and straddle-off the first package container **11m-4**; and carry **8-leg-2** the remaining second package container **11m-5** to the third location **7-loc-3** and straddle-off the second package container.

[0257] One embodiment is a system operative to autonomously deliver packages using a plurality of package containers, comprising: at least a first **11m-4** (FIG. **19C**) and a second **11m-5** (FIG. **19D**) package containers, in which the package containers are operative to contain packages **11pkg**

(FIG. 19C, FIG. 19D) to be delivered; and an on-road vehicle **10** (FIG. 19A, FIG. 19B) operative to straddle over the package containers **11m-4**, **11m-5**.

[0258] In one embodiment, the on-road vehicle **10** is configured to: straddle-over and grab the first package container **11m-4** parked at a warehouse location **7-loc-0** (FIG. 19F), as illustrated in FIG. 19A; straddle-over and grab the second package container **11m-5** parked at the same warehouse location **7-loc-0**, thereby grabbing both the first and second package containers **11m-4**, **11m-5** at the same time and in tandem, as illustrated in FIG. 19B; carry **8-leg-1** (FIG. 19F) the first and second package containers together **11m-4**, **11m-5** to a first location **7-loc-1** (FIG. 19F) and straddle-off the first package container **11m-4**, thereby leaving the first package container parked at the first location so as to allow pick-up of a first package **11pkg-41** (FIG. 19C); and carry **8-leg-2** (FIG. 19F) the remaining second package container **11m-5** to a third location **7-loc-3** (FIG. 19F) and straddle-off the second package container, thereby leaving the second package container parked at the third location so as to allow pick-up of a second package **11pkg-51** (FIG. 19D).

[0259] In one embodiment, the on-road vehicle **10** is further configured to: straddle-over grab and carry the first package container **11m-4** from the first location **7-loc-1** to a second location **7-loc-2**; and straddle-off the first package container **11m-4**, thereby leaving the first package container parked at the second location **7-loc-2** so as to allow pick-up of a third package **11pkg-42**.

[0260] In one embodiment, the on-road vehicle **10** is further configured to: straddle-over grab and carry the second package container **11m-5** from the third location **7-loc-3** to a fourth location **7-loc-4**; and straddle-off the second package container **11m-5**, thereby leaving the second package container parked at the fourth location **7-loc-4** so as to allow pick-up of a fourth package **11pkg-52**.

[0261] In one embodiment, the on-road vehicle **10** is further configured to: straddle-over grab and carry **8-leg-7** the second package container **11m-5** from a fourth location **7-loc-4** to a second location **7-loc-2**; and straddle-over and grab the first package container **11m-4** parked at the second location **7-loc-2**, thereby grabbing both the first and second package containers **11m-4**, **11m-5** at the same time and in tandem; and carry **8-leg-8** the first and second package containers together **11m-4**, **11m-5** to the warehouse location **7-loc-0** and straddle-off the first and second package container, thereby returning the first and second package containers to the warehouse location.

[0262] In one embodiment, at least one of the package containers **11m-4**, **11m-5** is a vending machine.

[0263] FIG. 20 illustrates one embodiment of a method for autonomously delivering packages using a plurality of package containers. In conjunction with receiving a plurality of indications associated respectively with a plurality of packages **11pkg**, in which each of the indications is an indication pertaining to a delivery status associated with a respective one of the packages, e.g., **11pkg-41**, in a respective one of a plurality of lockers, e.g., **11L-41**, of a respective one of the package containers, e.g., **11m-4**, that is parked in a current respective place, e.g., **7-loc-1**, the method includes (per each of the indication received):

[0264] In step **1121**: straddling-over and grabbing, by an on-road vehicle **10**, the respective package container, e.g., **11m-4**, from the current respective place, e.g., location **7-loc-1**.

[0265] In step **1122**: releasing and straddling-off, by the on-road vehicle **10**, said respective package container **11m-4** at a different respective place, e.g., **7-loc-2**.

[0266] In step **1123** sending an indication that said respective package container **11m-4** is now and/or will be in the different respective place **7-loc-2** for another of the packages, e.g., **11pkg-42**, to be picked-up from a different one of the lockers, e.g., **11L-42**, of said respective package container **11m-4**.

[0267] FIG. 21A illustrates one embodiment of an autonomous vehicle **10** featuring a hybrid locomotion system configured for enhanced mobility and load handling. The vehicle **10** comprises a main body **3-body**, which may house control systems, power sources, and potentially serve as a support structure. Connected to the main body **3-body** are a plurality of articulated legs, such as legs **2a-1**, **2a-2**, **2c-1**, **2c-2**, **2d-1**, and **2d-2**. These legs are actuated via respective joints, such as

joints **2a-j1**, **2a-j2**, **2c-j2**, and **2d-j2**, allowing for complex movements and height adjustments. Wheels **1a**, **1b**, **1c**, and **1d** are attached to the distal ends of respective articulated legs, providing efficient locomotion over suitable ground surfaces. The vehicle **10** may further include sensors **4**-sense for perceiving the environment, actuators such as actuator **6d** associated with the wheels or joints, and a connector mechanism **5-cnct** integrated with or positioned relative to the main body **3**-body for interacting with loads.

[0268] FIG. **21B** illustrates one embodiment of the autonomous vehicle **10** engaging with a load **11**. In this depiction, the system actuates the articulated legs (e.g., **2a-1**, **2a-2**) via their respective joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**) to modify the vehicle's stance relative to the ground (e.g., in a bending-like motion). This adjustment configures the vehicle **10** such that the main body **3**-body is lowered and positioned right over the load **11**. This demonstrates the capability of the system to straddle over a load **11** and come in contact therewith, as a preliminary step before lifting or securing the load for transport.

[0269] FIG. **21C** illustrates one embodiment of the autonomous vehicle **10** subsequent to engaging with and securing the load **11**, potentially following the straddling configuration shown in FIG. **21B**. After positioning and lowering itself over the load **11**, the system is configured to actuate a connector mechanism (such as **5-cnct** shown in FIG. **21A**, or another mechanism associated with the main body **3**-body) to grab the load **11**. Following the grabbing action, the system actuates the articulated legs (e.g., **2a-1**, **2a-2**) and joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**) to lift the load **11** off the ground surface, thereby securing it underneath the main body **3**-body. The figure shows the vehicle **10** transporting the now-lifted load **11**, maintaining it at a specific elevation range **11-ele** relative to the ground surface. This variable ground clearance, provided by the articulated legs and joints, enables safe transport of the under-body load **11**. This under-body transport configuration leverages the integrated capabilities of grabbing, lifting via leg actuation, and maintaining variable height afforded by the articulated legged-wheeled structure.

[0270] FIG. **22** illustrates one embodiment of a manipulation capability of the autonomous vehicle **10**, demonstrating interaction with objects at elevated heights. To achieve the necessary reach, the system is configured to actuate its rear articulated legs to adopt a stabilized, upright or elevated stance, primarily supported by these rear legs and their respective wheels. While maintaining this elevated stance, the vehicle **10** utilizes its front articulated legs, configured to act as manipulators or arms, to interact with a load **11**. In this specific embodiment, the front articulated legs are shown positioning the load **11** onto an elevated external surface, depicted as a shelf structure **110-self**. This capability highlights the system's versatility, extending beyond ground-level operations and under-body transport to include complex interactions such as retrieving or placing items on shelves or other raised platforms by dynamically altering its posture and using specific legs for manipulation.

[0271] In one embodiment demonstrating advanced manipulation, particularly for handling loads (**11**) requiring precise orientation, the system (**10**) utilizes a specialized posture and coordinated wheel control. The system first actuates its rear articulated legs to achieve a stable, elevated stance described previously (FIG. **22**). The front articulated legs are then positioned such that the front wheels (**1c**, **1d**) make direct contact with the load (**11**), acting as primary gripping or contact points. Instead of or in addition to using dedicated grippers, the system relies on the friction of the wheels (**1c**, **1d**) against the load surface, possibly augmented by pressure applied by the front legs.

Critically, as the system manipulates the load (**11**) using its front legs—for example, lifting it from/to an elevated surface (**110-shelf**), repositioning it, or preparing to place it underneath the main body (**3**-body)—the control sub-system (**8**, FIG. **26**) actively rotates the front wheels (**1c**, **1d**). This wheel rotation compensates for the changes in the front leg joint angles (e.g., **2c-j2**) during the manipulation, thereby ensuring the load (**11**) maintains a desired orientation, such as remaining substantially horizontal, throughout the posture change and movement sequence. This coordinated control of front leg joints and front wheel rotation enables delicate handling and stable orientation control of the grasped load.

[0272] FIG. 23 illustrates one embodiment of enhanced navigational capabilities of the autonomous vehicle **10** enabled by its hybrid locomotion system. The vehicle **10**, while carrying a load **11** underneath its main body, is depicted successfully navigating a significant obstacle, specifically stairs **111**-stairs. The coordinated use of the articulated legs and wheels allows the vehicle **10** to traverse complex terrain that would typically be inaccessible to purely wheeled autonomous systems, demonstrating a key advantage for applications requiring delivery or operation in varied environments.

[0273] One embodiment is a system (**10**) operative to deliver loads autonomously comprising: a main body (**3**-body) configured to support a load (**11**) positioned underneath said main body; a plurality of articulated legs (e.g., **2a-1**, **2a-2**, **2c-1**, **2c-2**, **2d-1**, **2d-2**) comprising joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**, **2d-j2**), connected to the main body (**3**-body); and a plurality of wheels (e.g., **1a**, **1b**, **1c**, **1d**), each wheel attached to a respective one of the articulated legs; wherein the articulated legs and joints are configured to control elevation of the main body (**3**-body); and wherein the system is configured to: actuate the wheels (**1a**, **1b**, **1c**, **1d**) to achieve forward movement of the load (**11**) supported underneath the main body (**3**-body); and further actuate the articulated legs (e.g., **2a-1**, **2a-2**) via the joints (e.g., **2a-j1**, **2a-j2**) to adjust said elevation for at least one of maintaining ground clearance **11**-ele of the load when navigating obstacles (e.g., stairs **111**-stairs in FIG. 23) and manipulating the load (**11**) relative to a ground surface. This system architecture, visualized generally in FIG. 21A and in operation in FIGS. 21B, 21C, and 23, combines the efficiency of wheeled locomotion for travel over relatively flat surfaces with the adaptability of articulated legs for adjusting height, handling loads, and traversing challenging terrain, all while carrying the payload underneath the main structure.

[0274] In one embodiment, manipulating the load (**11**) relative to the ground surface comprises actuating the articulated legs (e.g., **2a-1**, **2a-2**) via the joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**) to lower the main body (**3**-body) and release the load (**11**) onto the ground surface. For example, after transporting the load as shown in FIG. 21C, the system would perform the reverse operation of lifting, using the legs and joints to carefully decrease the elevation (**11**-ele) until the load (**11**) rests securely on the ground, after which a connector mechanism would release it.

[0275] In one embodiment, manipulating the load (**11**) relative to the ground surface comprises actuating the articulated legs (e.g., **2a-1**, **2a-2**) via the joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**) to lift the load (**11**) from the ground surface underneath the main body (**3**-body). This action is depicted in the transition from a configuration like FIG. 21B (where the vehicle is positioned over the load) to the transport configuration shown in FIG. 21C (where the load is elevated **11**-ele). The legs extend or reconfigure using the joints to increase the ground clearance, lifting the secured load.

[0276] In one embodiment, prior to said lifting of the load (**11**), the system is further configured to straddle over the load (**11**). As illustrated in FIG. 21B, the vehicle (**10**) utilizes its articulated legs (e.g., **2a-1**, **2a-2**) and overall structure to position its main body (**3**-body) directly above the load (**11**) while the load is resting on the ground, ensuring proper alignment for subsequent grabbing and lifting operations.

[0277] In one embodiment, navigating obstacles comprises actuating the articulated legs (e.g., **2a-1**, **2a-2**) via the joints (e.g., **2a-j1**, **2a-j2**) to navigate stairs (**111**-stairs). FIG. 23 explicitly shows the vehicle (**10**) using its legs to ascend or descend stairs (**111**-stairs) while carrying the load (**11**), demonstrating a key advantage over purely wheeled systems.

[0278] In one embodiment, a control sub-system (like processor **8** in FIG. 1D) is configured to independently actuate each articulated leg (e.g., **2a-1**, **2c-1**, **2d-1**) and its respective joint/s (e.g., **2a-j1**, **2c-j2**, **2d-j2**) to maintain a substantially horizontal orientation of the main body (**3**-body), and consequently the load (**11**) supported thereunder, while traversing uneven ground surfaces or obstacles. This active stabilization ensures the load remains level and secure even when the wheels (**1a**, **1b**, **1c**, **1d**) are on different height levels, as might occur on rough terrain or slopes.

[0279] In one embodiment, maintaining the substantially horizontal orientation occurs while

navigating stairs (**111**-stairs), including situations where front ones of the articulated legs (e.g., **2c**, **2d**) and rear ones of the articulated legs (**2a** and the other rear leg not numbered) are simultaneously in contact with different steps of said stairs (**111**-stairs). As shown conceptually in FIG. **23**, the independent control allows the front legs to be on a higher step while the rear legs are on a lower step (or vice versa), with the joints actuating to keep the main body (**3**-body) and load (**11**) level during the transition between steps.

[0280] In one embodiment, the load is a package (represented conceptually by **11**) to be delivered to a designated location. This specifies one application for the system, where the load (**11**) contains goods intended for delivery.

[0281] In one embodiment, the system further comprising sensors (**4**-sense) configured to perceive the ground surface, wherein the system utilizes data from said sensors (**4**-sense) to coordinate actuation of the wheels (**1a**, **1b**, **1c**, **1d**) and joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**, **2d-j2**). As indicated in FIG. **21A**, sensors (**4**-sense), which may include cameras, LiDAR, inertial measurement units, or other types, provide environmental awareness enabling the control sub-system to make informed decisions for navigation, obstacle avoidance, and stabilization.

[0282] In one embodiment, the system is further configured to actuate a first subset of said articulated legs, designated as rear legs (implied rear legs in FIG. **22**), and their respective joints to stabilize the main body (**3**-body) in an elevated stance, primarily supported by said rear legs and their wheels, thereby increasing a reach capability of a second subset of said articulated legs, designated as front legs (e.g., legs used for manipulation in FIG. **22**). FIG. **22** illustrates this concept, where the vehicle (**10**) adopts an upright posture, supported by its rear limbs, allowing the front limbs greater freedom and height for interaction tasks.

[0283] In one embodiment, the system is further configured to utilize said front legs and their respective joints as manipulators, akin to arms or hands, to grasp and retrieve the load (**11**) from an elevated surface, such as a shelf (**110**-self), while the main body (**3**-body) is maintained in said elevated stance by the rear legs. FIG. **22** explicitly depicts the front legs of the vehicle (**10**) acting as manipulators to handle the load (**11**) relative to the shelf (**110**-self), demonstrating this advanced capability enabled by the elevated stance.

[0284] In one embodiment, the system is further configured to subsequently actuate said front legs, acting as manipulators, to either position the load (**11**) on ground or attach the load (**11**) underneath said main body (**3**-body) for transport. After retrieving the load from the shelf (**110**-self) as shown in FIG. **22**, the front legs can then maneuver the load either to be placed directly on the ground or secured to the underside of the main body (**3**-body) for subsequent transport, similar to the configuration in FIG. **21C**.

[0285] In one embodiment, the system further comprising at least one connector mechanism (e.g., **5**-cnct in FIG. **21A**) associated with the main body (**3**-body), wherein said connector mechanism is configured to secure the load (**11**) underneath the main body (**3**-body) during transport. This mechanism ensures the load remains firmly attached to the vehicle while it is moving, particularly when lifted as in FIG. **21C**.

[0286] In one embodiment, the connector mechanism is configured to actively grab the load (**11**) prior to the system starting manipulating the load (**11**) to lift it from the ground surface. Before the legs actuate to lift the load (as in the transition from FIG. **21B** to **21C**), the connector mechanism engages and secures the load.

[0287] In one embodiment, the connector mechanism is configured to release the load (**11**) after the system has finished manipulating the load (**11**) to lower it onto the ground surface. Once the legs have lowered the load (reverse of lifting from FIG. **21C**), the connector mechanism disengages, allowing the vehicle to move away.

[0288] In one embodiment, the connector mechanism comprises at least one of: a. an electromagnetic connector; b. a mechanical gripper or clamp; c. a vacuum suction cup; and d. a hook-and-latch mechanism. These represent various technologies that could be implemented for the

connector mechanism (e.g., 5-cnct) to achieve the grabbing and releasing functions depending on the nature of the load and the operational requirements.

[0289] In one embodiment, the load (11) comprises a complementary connector interface configured to engage with the connector mechanism (e.g., 5-cnct) of the autonomous vehicle system (10). For reliable connection, the load (11) itself would possess features (e.g., a ferromagnetic plate for an electromagnet, specific geometry for grippers, a suitable surface for suction) designed to mate with the vehicle's connector mechanism.

[0290] FIG. 24 illustrates one embodiment of a method for delivering a load autonomously using a system comprising a main body (3-body), articulated legs (e.g., 2a-1) with joints (e.g., 2a-j1) connected to the main body, and wheels (e.g., 1a) attached to the articulated legs, the method comprising: securing the load (11) underneath the main body (3-body) of the autonomous system (step 1131); actuating the wheels (e.g., 1a, 1c) to achieve forward movement, thereby transporting the secured load (11) towards a designated delivery location over a ground surface (step 1132); upon arrival at or near the designated delivery location, at least reducing said forward movement achieved by actuating the wheels (step 1133); and actuating the articulated legs (e.g., 2a-1, 2c-1) via the joints (e.g., 2a-j1, 2c-j2) to manipulate the load (11) relative to a ground surface at the designated delivery location, thereby delivering the load (step 1134). The initial securing (1131) might follow a lift operation. The transport phase (1132) primarily utilizes the efficiency of wheeled motion. Upon reaching the destination, the system slows down or stops (1133) before using the adaptability of the legs and joints (1134) to perform the final delivery manipulation, such as lowering and releasing the load. The entire process may be orchestrated by a control sub-system (8).

[0291] FIG. 25 illustrates one embodiment a method for autonomously transporting a load using a system comprising a main body (3-body) supporting the load (11) underneath, articulated legs (e.g., 2a-1) with joints (e.g., 2a-j1) connected to the main body, and wheels (e.g., 1a) attached to the articulated legs, the method comprising: securing the load (11) underneath the main body (3-body) of the autonomous system (step 1141); actuating the wheels (e.g., 1a, 1c) to achieve forward movement, thereby transporting the secured load (11) towards a target location over a ground surface (step 1142); and dynamically actuating the articulated legs (e.g., 2a-1, 2c-1) via the joints (e.g., 2a-j1, 2c-j2) to maintain a ground clearance (e.g., 11-ele in FIG. 21C) between the load (11) supported underneath the main body (3-body) and the ground surface while traversing variations in the ground surface and/or encountering obstacles (step 1143). This method emphasizes the active ride height control during transport. While the wheels (controlled by actuators 6) provide primary propulsion (1142), the control sub-system (8), using inputs from sensors (4), simultaneously commands the legs/joints (actuators 2) to make continuous adjustments (1143) ensuring the under-body load (11) does not collide with the ground or low obstacles.

[0292] In one embodiment, maintaining said ground clearance includes increasing elevation of the main body (3-body) when an obstacle is detected. As the system transports the load using wheels, sensors (4) may detect a curb, rock, stairs, or other obstacle. The control sub-system (8) processes this information and commands the articulated legs and joints (2) to actuate, maintaining or temporarily increasing the ground clearance (11-ele in FIG. 21C) to allow the vehicle and the under-body load (11) to pass safely over the obstacle without interrupting forward movement significantly.

[0293] In one embodiment, maintaining said ground clearance includes decreasing elevation of the main body (3-body) when traversing a relatively smooth and level portion of the ground surface to improve stability and/or lower the center of gravity. During transport over flat terrain, where maximum ground clearance is not needed, the control sub-system (8) may actuate the legs and joints (2) to slightly lower the main body (3-body) and load (11). This lowers the overall center of gravity, enhancing stability during wheeled motion and potentially reducing energy consumption or wind resistance.

[0294] In one embodiment, traversing variations in the ground surface includes traversing uneven terrain and/or stairs (**111-stairs**). The dynamic actuation of legs and joints to maintain ground clearance is particularly crucial when navigating challenging surfaces such as rough, uneven ground or ascending/descending stairs (**111-stairs**), as depicted in FIG. **23**. The system continuously adjusts leg positions and joint angles to keep the load (**11**) clear of the steps or ground variations while the wheels (e.g., **1a**, **1c**) provide traction or support as needed.

[0295] In one embodiment, maintaining of ground clearance within a predetermined range involves specific numerical targets adjusted dynamically based on terrain and obstacles, as perceived by sensors (**4**) and processed by the control sub-system (**8**). For instance, during transport over a smooth, level ground surface, the system might maintain a nominal ground clearance (**11-ele** in FIG. **21C**) between the lowest point of the load (**11**) and the ground within a relatively low range, perhaps 5 to 10 centimeters, to optimize stability by keeping the center of gravity low. Upon detecting an upcoming obstacle, such as a standard curb of 15 centimeters, the control sub-system (**8**) would proactively command the articulated legs and joints (actuators **2**, **6**) to increase the ground clearance to 20 to 25 centimeters just before reaching the obstacle, ensuring the load (**11**) passes safely over it. When traversing continuously uneven terrain or stairs (**111-stairs**), the dynamic adjustment of ground clearance becomes even more critical. For stairs specifically (FIG. **23**), as the front wheels approach the first step, the front legs may lift to place the wheels onto the step while simultaneously adjusting joint angles to begin raising the main body (**3-body**). Throughout this maneuver, where front and rear legs are often on different steps, the system continuously adjusts the height of all legs to maintain the load (**11**) at a safe clearance, perhaps 5-10 centimeters above the edge (nose) of each stair step it is currently passing over, while also potentially maintaining the main body and load in a substantially horizontal orientation. The specific values within these clearance ranges are predetermined based on the vehicle's (**10**) limits, load (**11**) characteristics, and the operating environment, and are continuously monitored and adjusted.

[0296] FIG. **26** illustrates one embodiment of a control sub-system **8** configured to manage the autonomous operation of the vehicle **10**. The control sub-system **8** receives input data from various sensors **4** (e.g., corresponding to **4-sense** in FIG. **21A**), which perceive the external environment and internal vehicle state. Within the control sub-system **8**, processing resources include memory **9-mem**, a central processing unit (CPU) **9-CPU**, and potentially a graphics processing unit (GPU) **9-GPU**, which is often beneficial for parallel processing tasks common in robotics and machine learning. These resources execute control logic and algorithms, including, in this embodiment, a machine learning (ML) model **9model**. The ML model **9model** may be trained to perform complex tasks such as object recognition, terrain classification, path planning, gait generation for the articulated legs, or determining optimal coordination between wheels and legs based on the sensor **4** inputs. The outputs generated by the control sub-system **8**, derived from processing sensor data potentially using the ML model **9model**, are command signals sent to actuators **2** and **6**. These actuators correspond to the motors and mechanisms controlling the articulated legs, joints, and wheels of the vehicle **10**, thereby translating computational decisions into physical movement and manipulation actions of the autonomous system.

[0297] In one embodiment, the adaptable legged-wheeled locomotion system described herein (e.g., **10** in FIG. **21A**) can be scaled to address the diverse load types and operational environments depicted in FIG. **16**. A large-scale version, analogous to vehicle **10-large**, could transport substantial loads (**11-large**) exceeding 2 meters in dimension, such as specialized equipment containers or building materials. Its articulated legs would enable navigation over rough construction sites or uneven loading docks inaccessible to standard wheeled platforms, and allow precise positioning or straddling over ground-level obstacles during handling. A medium-scale implementation, similar to **10-medium**, carrying loads like furniture, appliance boxes, or potentially adapted passenger cabins (**11-medium**, approx. 1-2 meters), could leverage its legs to climb

entrance stairs (like **111**-stairs in FIG. 23), delivering bulky items directly inside buildings. A small-scale version, akin to **10**-small, would excel at last-meter delivery, carrying parcels or grocery totes (**11**-small, approx. 0.5-1 meter) from a curbside point, autonomously navigating sidewalks, ascending porch steps, and using leg manipulation to place the package securely. Finally, a tiny-scale variant (**10**-tiny) carrying small, critical items like documents or medical supplies (**11**-tiny, <0.5 meters) could navigate complex indoor environments, overcoming thresholds, small steps, or clutter found in offices or hospitals. Across all scales, the core benefit remains: combining efficient wheeled travel with legged adaptability for obstacle negotiation and load manipulation appropriate to the size and nature of the cargo.

[0298] Exemplary Scenario 1: Curbside to Apartment Door Delivery. In one delivery scenario, the autonomous vehicle (**10**) might be deployed from a larger delivery van parked at the curb. Initially, it would actuate its wheels (e.g., **1a**, **1c**) to efficiently traverse the sidewalk. Upon encountering a curb or uneven section, the control sub-system (**8**), using input from sensors (**4**), would actuate the articulated legs and joints to increase ground clearance (**11**-ele), stepping over the obstacle while maintaining stability and load orientation (claim **1.3b**). If the destination involves navigating entryway stairs (**111**-stairs, FIG. 23), the system transitions to primarily using its articulated legs, independently actuating each leg (claim **1.3b**) to ascend the steps while keeping the under-body package level and clear of the stair edges. Finally, upon reaching the recipient's door (the designated location), the vehicle stops wheeled motion and actuates the legs and joints to lower the main body (**3**-body) and release the package (**11**) onto the floor or doorstep.

[0299] Exemplary Scenario 2: Multi-Story Building Delivery via Elevator. For deliveries within a multi-story building, the autonomous vehicle (**10**) would first navigate the lobby using its wheels. Upon reaching an elevator bank, sensors (**4**) would identify the call button. The system would then actuate its rear legs to achieve a stable, elevated stance. Utilizing its front legs as manipulators, it would extend one leg to press the elevator call button. Once the elevator arrives and the doors open, the vehicle (**10**) might need to temporarily lower its stance slightly or adjust its legs to enter. To safely ride the elevator while potentially needing its legs for stability or height adjustment, the system could be configured to place the load (**11**) onto the elevator floor. During the ride, it might maintain the elevated stance for stability. After exiting the elevator, it would retrieve the load (**11**) back underneath its main body (**3**-body), resume wheeled locomotion to the recipient's office or apartment door, and finally deliver the load by lowering and releasing it.

[0300] Exemplary Scenario 3: Warehouse Shelf Retrieval and Loading. In a warehouse or fulfillment center setting, the autonomous vehicle (**10**) can automate the initial loading process. It would navigate aisles using its wheels. Upon reaching a designated shelf (similar to **110**-self in FIG. 22), the system actuates its rear legs to adopt the elevated stance (claim **1.6**), increasing the reach of its front legs. The front legs, acting as manipulators would then grasp a specific package or load (**11**) from the shelf. Once the load is securely grasped, the system actuates the front legs again to maneuver the load underneath the main body (**3**-body) and secure it using a connector mechanism. The vehicle (**10**) is then ready to transport the load using its wheels.

[0301] Exemplary Scenario 4: Uneven Terrain Site Delivery. Consider a scenario delivering supplies or tools (load **11**) on a construction site or across uneven outdoor terrain. The autonomous vehicle (**10**) would primarily use its wheels on flat sections. However, when encountering rough patches, debris, or significant slopes, the control sub-system (**8**) would rely heavily on dynamically actuating the articulated legs and joints to maintain a safe ground clearance (**11**-ele) and stability. The system would continuously adjust leg height and joint angles to keep the main body (**3**-body) and load (**11**) level while stepping over rocks or navigating muddy sections, ensuring the load arrives safely at the designated work zone where it can be lowered and released.

[0302] Exemplary Scenario 5: In a common urban or suburban delivery scenario, the autonomous vehicle (**10**), carrying a package secured underneath its main body (**3**-body), might start its journey by navigating public roads alongside regular vehicle traffic. During this phase, the system primarily

actuates its wheels for efficient forward movement, relying heavily on its sensors and control sub-system (8) for lane keeping, speed control, and collision avoidance relative to other vehicles. Upon approaching the designated delivery address, the system identifies a suitable location to transition off the road, such as a driveway entrance or a clear section of curb. It then carefully actuates its articulated legs and joints to mount the curb, dynamically adjusting ground clearance (11-ele) to ensure the load (11) clears the elevation change while maintaining stability. Once on the sidewalk or pathway, it may resume primarily wheeled locomotion towards the building entrance. If the final approach involves navigating stairs (111-stairs, FIG. 23), the system again transitions its locomotion strategy. It ceases significant forward wheel motion and predominantly uses the articulated legs and joints to ascend the steps, independently actuating each leg to maintain a level orientation for the main body and load and ensuring safe clearance over each stair edge. Having successfully navigated the stairs, the vehicle reaches the doorstep (final designated location), stops, and performs the final manipulation by actuating the legs and joints to lower and release the package (11).

[0303] In one embodiment, in preparation for manipulating a load (11), such as performing a straddle-and-lift operation or retrieving an item from a shelf, the autonomous system (10) employs its sensors (4) to first perceive the target load. The control sub-system (8), potentially utilizing image processing algorithms or machine learning models (9model, FIG. 26), analyzes the sensor data to gauge the size and estimate the dimensions of the load (11). Based on these determined dimensions, the system preemptively adjusts the variable ground clearance (11-ele, FIG. 21C) by actuating the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1). This proactive adjustment ensures sufficient vertical and lateral space is created underneath the main body (3-body) before the vehicle attempts to position itself over or grasp the load, preventing potential collisions. Furthermore, the control sub-system (8) may also adjust actuator settings for the legs and joints, such as movement speed, torque limits, or joint stiffness, based on the perceived size or potentially estimated weight of the load (11) to ensure safe and appropriate handling during manipulation (claim 1.0).

[0304] In one embodiment, prior to manipulating a load (11), particularly for operations like straddling over it or lifting it, the autonomous system (10) utilizes its sensors (4) to perceive the load. The control sub-system (8) processes the sensor data to specifically determine the height or vertical dimension of the load (11) relative to the ground surface. Based on this perceived load height, the control sub-system (8) calculates the required minimum elevation for the main body (3-body) to ensure adequate clearance. The system then actuates the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) to adjust the main body's elevation (affecting 11-ele, FIG. 21C) to meet or exceed this calculated height. This proactive height adjustment based on perceived load dimensions prevents collision between the main body or connector mechanism and the load itself during the approach and manipulation phases.

[0305] In one embodiment, during forward movement utilizing the wheels, the system continuously monitors the upcoming terrain using its sensors (4). The control sub-system (8) processes this sensor data, potentially using the ML model (9model), to determine the height profile of the ground surface and detect the height of any potential obstacles in the vehicle's path. This real-time height information is crucial for the dynamic actuation of the articulated legs and joints. Based on the perceived height of the upcoming ground or obstacles relative to the current elevation of the load (11) supported underneath the main body (3-body), the control sub-system (8) calculates and commands precise adjustments to the articulated legs and joints. This ensures that the variable ground clearance (11-ele, FIG. 21C) is constantly maintained within the predetermined safe range (claim 3.0), proactively increasing clearance to step over raised obstacles (claim 3.1) or accommodating dips and variations in the terrain, thereby preventing the under-body load (11) from impacting the ground during transport.

[0306] In one embodiment, before initiating or during autonomous transport, the system (10) may utilize its sensors (4) not only to perceive the ground but also to gauge or confirm the height

(vertical dimension) of the load (11) itself before it has been secured underneath the main body (3-body). This measured load height is critical input for the control sub-system (8). To ensure safe transport, the control sub-system determines a target minimum ground clearance required between the bottom of the load (11) and the ground surface (e.g., the lower bound of the predetermined safe range). Knowing the load's height, the control sub-system calculates the necessary minimum elevation for the main body (3-body) relative to the ground that will achieve this target load-to-ground clearance. It then actuates the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) to position the main body (3-body) at or above this calculated minimum elevation, establishing the desired initial height (representing the clearance under the body, which indirectly sets clearance under the load) before or as wheeled motion commences. This ensures the load (11) starts its journey with sufficient separation from the ground, based directly on its own measured dimensions.

[0307] In one embodiment, in certain locomotion scenarios where precise foot placement and stability are paramount, such as when navigating obstacles like stairs (111-stairs, FIG. 23), the control sub-system (8) may be configured to operate the system (10) in a predominantly legged walking gait, temporarily deviating from primary reliance on wheeled motion. During this legged gait, the control sub-system (8) may command actuators (2, 6, FIG. 26) associated with the wheels (e.g., 1a, 1b, 1c, 1d) to actively lock or apply brakes to prevent unintended rolling. By locking the wheels when a specific leg is being used as a stationary support point (akin to a foot placement), the system enhances stability and positional accuracy. This allows the articulated legs (e.g., 2a-1, 2c-1) and joints (e.g., 2a-j1, 2c-j2) to execute precise movements for lifting the body, stepping onto the next stair, and maintaining balance, ensuring safe and controlled traversal of complex terrain where wheeled rolling would be ineffective or detrimental. Once back on a suitable surface, the wheel locks can be released to resume efficient wheeled locomotion.

[0308] In one embodiment, the autonomous system (10), particularly in its legged-wheeled configuration (FIG. 21A), is designed with versatility to handle a wide range of loads (11), adapting to diverse delivery and transport needs. As explicitly shown, the system can transport generic boxed loads (11 in FIGS. 21B, 21C, 23, 22). This load (11) may represent a standard package to be delivered. Furthermore, drawing context from related embodiments, the load (11) handled by the legged-wheeled vehicle (10) can encompass more specialized items. These include detachable containers (e.g., analogous to 11m-4, 11m-5 in FIGS. 19C, 19D) which may themselves contain multiple lockers (e.g., 11L-41, 11L-51) for secure, multi-recipient deliveries. The system can also be configured to handle functional loads (e.g., analogous to 11-func in FIG. 7B), such as portable power sources, fuel tanks, or communication equipment. Additionally, adapting concepts from FIGS. 17A-17D, the system, potentially via a specialized connector or gripper, could be configured to handle disposable packages, such as cardboard boxes (11-crb), plastic bags (11-bag), or items secured with straps (92-strap), especially for last-meter delivery scenarios where the legged mobility is advantageous for reaching doorsteps or navigating complex pathways after initial transport by another system. The ability to perceive the load's dimensions allows the system to adjust its handling strategy for these varied load types.

[0309] In one embodiment, subsequent to lifting the load (11) underneath the main body (3-body), the system (10) provides enhanced load securing through active leg actuation. The control sub-system (8, FIG. 26) may be configured to command multiple articulated legs via their respective joints to contract inwards towards the secured load (11). This inward motion applies controlled pressure, effectively clamping or bracing the load (11). This leg-based securing mechanism can complement or form an integral part of a dedicated connector mechanism (5-cnct). Beyond providing stability during transport over uneven terrain or stairs (111-stairs, FIG. 23) and physical protection, this inward leg contraction also serves as an active anti-theft measure. The control sub-system (8), using input from sensors (4, 4-sense) such as force sensors, proximity sensors, or cameras, can be configured to detect an attempt at unauthorized removal of the load (11). Upon detecting such an attempt, the system can increase the clamping force applied by the articulated

legs, making it significantly more difficult to forcibly detach the load (**11**) from the vehicle (**10**), thereby deterring theft while the load is in transit or temporarily placed.

[0310] In one embodiment, while the embodiments illustrated herein, particularly in FIGS. **21A-23**, depict an autonomous vehicle (**10**) with four articulated legs, each equipped with a wheel, this configuration is presented by way of example and not limitation. The fundamental principles of the system—combining articulated legs and wheels for hybrid locomotion, providing variable ground clearance for under-body load transport, and utilizing legs for manipulation and obstacle negotiation—can be implemented using alternative leg configurations. Therefore, the scope of the invention encompasses autonomous vehicles utilizing a different plurality of articulated legs, such as three, six, or potentially more legs, depending on specific design requirements for stability, payload capacity, redundancy, or maneuverability, while still incorporating wheels on at least some of the legs and retaining the core functionalities described.

[0311] In various embodiments, at least some of the joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**, **2d-j2** in FIG. **21A**) connecting the segments of the articulated legs (e.g., **2a-1**, **2a-2**) and connecting the legs to the main body (**3-body**) are configured to allow multiple degrees of freedom (DOF). For example, a proximal joint connecting a leg to the body might allow for rotation akin to a hip (providing forward/backward and side-to-side motion), while a distal joint might function like a knee or ankle (providing primarily pitching motion). Having multiple joints per leg, each potentially offering 2-DOF or 3-DOF (e.g., incorporating pitch, roll, and yaw axes), facilitates a large workspace and complex kinematics. This multi-DOF configuration enables the control sub-system (**8**) to orchestrate intricate movements, facilitating a full or partial humanoid-like motion span for the articulated legs. Examples include not only lifting the body vertically or stepping forward/backward but also enabling side-stepping, torso twisting relative to the legs, complex leg folding for compact storage, and the versatile manipulation capabilities shown in FIG. **22** where the front legs act akin to arms with shoulder, elbow, and wrist-like articulations. This enhanced range of motion is crucial for navigating highly complex terrain, maintaining stability across varied postures, and performing sophisticated load manipulation tasks.

[0312] In one embodiment, the system's capability for manipulating the load relative to a ground surface extends beyond simple lowering and releasing to achieve centimeter-level accuracy in spatial placement. This precision is enabled by the complex kinematics afforded by the multi-jointed articulated legs (e.g., **2a-1**, **2a-2**) and the sophisticated algorithms executed by the control sub-system (**8**, FIG. **26**). Utilizing feedback from high-resolution sensors (**4**, **4-sense**), including potentially proprioceptive sensors within the joints themselves, the control sub-system can orchestrate intricate, fine-tuned movements of the individual legs and joints. This allows the main body (**3-body**), and consequently the load (**11**) held beneath it or manipulated by front legs, to be positioned with high accuracy in three-dimensional space relative to the target delivery surface. This capability is crucial for scenarios requiring precise placement, such as depositing a package into a narrow designated receptacle, aligning the load with specific markers on the ground, or carefully setting down fragile items within tight tolerances, ensuring not just delivery but accurate and safe final positioning of the load.

[0313] In various embodiments, actuation of the system's moving components, including providing rotational force to the wheels (e.g., **1a**, **1b**, **1c**, **1d**) for locomotion and controlling the movement of the articulated legs (e.g., **2a-1**, **2a-2**) via their respective joints (e.g., **2a-j1**, **2a-j2**, **2c-j2**, **2d-j2**), is accomplished using one or more actuators (represented generally by **2**, **6** in FIG. **26**, and **6d** in FIG. **21A**). In various embodiments, these actuators may comprise electric motors, such as servo motors or brushless DC motors, known for their precise control and efficiency, often integrated near or within the joints. Alternatively, particularly for applications demanding high force or torque for lifting heavy loads or robust leg articulation, hydraulic actuation systems utilizing pressurized fluid and cylinders may be employed. Other technologies, including pneumatic systems or advanced materials-based actuators, could also be implemented depending on the specific scale, power,

speed, and precision requirements of the autonomous vehicle (10). The selection of the actuation method is a key design choice impacting the overall performance, capabilities, and operational characteristics of the system.

[0314] In one embodiment, the autonomous delivery system (10) offers capabilities beyond simple ground-level load release, enabling precision placement of packages (11) tailored to specific recipient needs or environmental conditions. Upon arriving at the designated delivery location and ceasing forward wheeled motion, the control sub-system (8) utilizes sensor data (4-sense,) to identify the precise target placement spot. This might be a slightly elevated porch bench to keep the package off wet ground, a specific designated area within a larger porch, or inside a customer-provided secure delivery receptacle. The system then finely actuates the articulated legs and joints not just to lower the load, but to carefully maneuver the main body (3-body) and potentially utilize the front legs as fine manipulators to position the package (11) at the exact target coordinates. Furthermore, for packages marked as fragile or requiring a specific orientation (e.g., 'This Side Up'), the system can control the final release from the connector mechanism and the leg movements to ensure the package is deposited gently and maintains the required orientation on the surface. This precision enhances package safety, security, and meets specific customer requirements.

[0315] In one embodiment, in a typical operational scenario, the autonomous vehicle (10) is loaded at a depot with multiple distinct packages (conceptually represented by load 11, but understood to be multiple individually addressable items secured underneath or within the main body 3-body), each destined for a different location along a pre-planned route. The control sub-system (8, FIG. 26) optimizes the delivery sequence. The vehicle (10) departs the depot, primarily utilizing its wheels for efficient travel along roads and sidewalks towards the first designated delivery location. During transit, it continuously uses its articulated legs and joints to maintain safe ground clearance (11-ele) over varying terrain and obstacles. Upon arriving near the first address, the vehicle may need to navigate specific obstacles like curbs or a few stairs (111-stairs, FIG. 23) using primarily its legs, potentially locking its wheels for stability and maintaining a level orientation. Reaching the precise drop-off point (e.g., the front porch), the vehicle stops, actuates its legs and joints to lower the main body, and releases only the first specific package using its connector mechanism. Having completed the first delivery, the system secures any remaining packages, adjusts its stance, and proceeds towards the next delivery location on its route, again primarily using wheeled locomotion, repeating the process of navigation, obstacle traversal (if needed), and precise package release until all packages for that route have been delivered.

[0316] In one embodiment, the powered actuation system as described herein may be embodied, for example, by the linear actuators (2'+2) enabling vertical movement of the wheel structures (2, 2') relative to the main body (3) in the embodiment of FIG. 1A, and/or by the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) enabling movement of the wheels (e.g., 1a) relative to the main body (3-body) in the embodiment of FIG. 21A, and/or by any other means.

[0317] One embodiment is an autonomous vehicle (e.g., 10 in FIG. 1A; 10 in FIG. 21A), comprising: a main body structure (e.g., upper horizontal structure 3 in FIG. 1A; main body 3-body in FIG. 21A); a plurality of wheels (e.g., 1, 1a, 1b, 1c, 1d in FIG. 1A; 1a, 1b, 1c, 1d in FIG. 21A); a plurality of powered actuation systems (e.g., linear actuators 2'+2 in FIG. 3C/3D; articulated legs 2a-1/joints 2a-j1 in FIG. 21A), wherein for each wheel (e.g., 1a) of the plurality of wheels, a respective one of the powered actuation systems (e.g., associated linear actuator 2a'/2a; associated leg 2a-1/joints 2a-j1) is configured to actively modify a relative positioning between said wheel (e.g., 1a) and the main body structure (e.g., 3; 3-body); and a grabbing mechanism (e.g., 5-cnct in FIG. 1A/3B; 5-cnct in FIG. 21A) coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load (e.g., 11 in FIG. 3D; 11 in FIG. 21C) beneath the main body structure and above ground; wherein the autonomous vehicle (10) is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning

between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load (e.g., **11**) relative to the ground.

[0318] In one embodiment, the load (e.g., **11**) comprises a consumer item (e.g., **93-ci** in FIG. **17A**) to be delivered to a drop-off location.

[0319] In one embodiment, the consumer item (**93-ci**) is contained within a disposable package (e.g., **11-crb** in FIG. **17A**; **11-bag** in FIG. **17D**), and wherein the grabbing mechanism (e.g., **5-cnct** in FIG. **17A/17B**; **5-side-arm** in FIG. **17C**) is configured to directly hold said disposable package.

[0320] In one embodiment, the dimensions of the autonomous vehicle (**10**) are configured to allow the vehicle to traverse a sidewalk prior to reaching the drop-off location.

[0321] In one embodiment, said dimensions comprise a length of less than 2.5 meters, a height of less than 1.5 meters, and a weight of less than 700 kilograms.

[0322] In one embodiment, the vehicle (**10**) is configured to traverse roads (e.g., **20a** in FIG. **2A**; **20c** in FIG. **4B**) prior to said sidewalk traversal. The vehicle is capable of operating on standard roadways alongside traffic before transitioning to sidewalks for final delivery approach.

[0323] In one embodiment, said dimensions comprise a length of less than 1.5 meters, a height of less than 1 meter, and a weight of less than 200 kilograms. These smaller dimensions are suitable for very compact vehicles potentially operating primarily in pedestrian areas or indoors.

[0324] In one embodiment, said manipulation of the load (**11**) relative to the ground comprises the vehicle (**10**) being configured to: control the plurality of powered actuation systems (e.g., **2'+2**; legs/joints) to coordinately modify said relative positioning to lower the load (**11**) towards the ground surface (e.g., transition from FIG. **3D** to **3C**; or from FIG. **21C** to **21B**); and operate the grabbing mechanism (e.g., **5-cnct**) to release the load (**11**) onto the ground surface.

[0325] In one embodiment, the autonomous vehicle is further configured to subsequent to the grabbing mechanism releasing the load (**11**), straddle off the load.

[0326] In one embodiment, said manipulation of the load (**11**) relative to the ground comprises the vehicle (**10**) being configured to: position the main body structure (e.g., **3**; **3-body**) over the load (**11**) at a pick-up location (e.g., FIG. **3B**; FIG. **21B**); operate the grabbing mechanism (e.g., **5-cnct**) to hold the load (**11**); and subsequently control the plurality of powered actuation systems (e.g., **2'+2**; legs/joints) to coordinately modify said relative positioning to lift the load (**11**) from the ground surface (e.g., transition from FIG. **3C** to **3D**; transition from FIG. **21B** to **21C**).

[0327] In one embodiment, said coordinated modification of said relative positioning, enabled by the control of the plurality of powered actuation systems (e.g., **2'+2**; legs/joints) to manipulate the load (**11**) relative to the ground, includes maintaining a clearance (e.g., **9-clr** in FIG. **1A**; **11-ele** in FIG. **21C**) between a bottom surface of the load (**11**) and a ground surface during transport of the load by the autonomous vehicle (**10**).

[0328] In one embodiment, said clearance (e.g., **9-clr**; **11-ele**) is at least 10 (ten) centimeters.

[0329] In one embodiment, said providing active suspension comprises each respective powered actuation system (e.g., **2'+2**; legs/joints) functioning as at least one of a spring and a damper between its associated wheel (e.g., **1a**) and the main body structure (e.g., **3**; **3-body**).

[0330] In one embodiment, said providing active suspension enables vertical movement of each said wheel (e.g., **1a**) relative to the main body structure (e.g., **3**; **3-body**) responsive to variations in the ground surface encountered during said forward motion. This describes the outcome of the active suspension—allowing wheels to move up and down independently to absorb bumps, curbs and stairs while the main body remains relatively stable.

[0331] In one embodiment, the autonomous vehicle further comprises a plurality of sensors (e.g., **4**, **4a**, **4b**, **4c** in FIG. **1A**; **4-sense** in FIG. **21A**; **4** in FIG. **26**) configured to collect data representing a surrounding environment of the autonomous vehicle (**10**).

[0332] In one embodiment, the autonomous vehicle (**10**) is further configured to generate, based on

the data collected by the plurality of sensors (e.g., **4**; **4-sense**), a real-time three-dimensional representation (e.g., via model **9model** in FIG. **26**) of the surrounding environment, said representation including the load (**11**) when present within the surrounding environment.

[0333] In one embodiment, said configuration to control the plurality of powered actuation systems (e.g., **2'+2**; legs/joints) to manipulate the load (**11**) relative to the ground is based at least in part on said three-dimensional representation (e.g., **9model**) of the surrounding environment. The vehicle uses its environmental understanding to accurately position itself and manipulate the load.

[0334] FIG. **27** illustrates one embodiment is a method for autonomously delivering a load (e.g., **11**), the method comprising: receiving, by a control system (e.g., **8** in FIG. **26**) of an autonomous vehicle (**10**) comprising a main body structure (e.g., **3**; **3-body**) and a plurality of wheels (e.g., **1**; **1a**), instructions associated with delivering the load to a drop-off location (step **1151**); transporting the load (**11**), held by a grabbing mechanism (e.g., **5-cnct**) coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels (e.g., step **1132** in FIG. **24**), wherein during said transporting, active suspension is provided by a plurality of powered actuation systems (e.g., **2'+2**; legs/joints), wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure (step **1152**); upon arrival at the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated modification of said relative positioning between each of the wheels and the main body structure to manipulate the load (**11**) relative to a ground surface (step **1153**); and operating the grabbing mechanism to release the load (step **1154**). This method claim outlines the delivery process utilizing the dual-purpose actuation systems.

[0335] FIG. **28** illustrates one embodiment of a method for autonomously delivering a load (e.g., **11**), the method comprising: receiving, by a control system (e.g., **8**) of an autonomous vehicle (**10**) comprising a main body structure (e.g., **3**; **3-body**) and a plurality of wheels (e.g., **1**; **1a**), instructions associated with delivering the load to a drop-off location (step **1161**); transporting the load (**11**), held by a grabbing mechanism (e.g., **5-cnct**) coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems (e.g., **2'+2**; legs/joints), wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure (step **1162**); and coordinately controlling, via the control system (**8**) during at least a portion of said transporting, said plurality of powered actuation systems (e.g., **2'+2**; legs/joints) to maintain a clearance (e.g., **9-clr**; **11-ele**) between a bottom surface of the load (**11**) and a ground surface (step **1163**).

[0336] In one embodiment, said clearance (e.g., **9-clr**; **11-ele**) is at least 10 (ten) centimeters.

[0337] In one embodiment, the autonomous vehicle (**10**) actively maintains a safe ground clearance (e.g., **9-clr**, **11-ele**) for the load (**11**) carried underneath its main body (**3**, **3-body**) using a closed-loop sensing and control process. Primarily, forward-looking sensors (**4**, **4-sense**, potentially LiDAR, cameras, radar) perceive the upcoming terrain and obstacle height profile. This data feeds into the control sub-system (**8**, FIG. **26**), which calculates and commands dynamic adjustments to the powered actuation systems (**2'+2** or articulated legs/joints via actuators **2**) to proactively modify the wheel-body relative positioning, thus ensuring adequate clearance before encountering variations. Additionally, the system may employ complementary sensing methods for reactive adjustments or verification. For instance, downward-facing visual sensors (part of **4** or **4-sense**) could directly monitor the actual clearance between the load's bottom surface and the ground in real-time. Furthermore, inertial measurement units (IMUs) within the sensor suite (**4**, **4-sense**) could detect sudden accelerations or jolts indicative of an unintended contact, while acoustic sensors might detect the sound of such contact. In some embodiments, mechanical contact sensors or force/torque sensors associated with the load (**11**) or the grabbing mechanism (**5-cnct**) could

provide direct feedback upon an initial, minor collision of the load's bottom with the ground. Upon detecting such contact via these secondary sensing methods, the control sub-system (8) would immediately trigger a corrective action, commanding the powered actuation systems to rapidly increase the ground clearance to prevent further impact or damage.

[0338] In one embodiment, the autonomous vehicle (10), irrespective of whether its powered actuation system comprises linear actuators (e.g., 2'+2 in FIG. 3C/3D) or articulated legs with joints (e.g., 2a-1, 2a-j1 in FIG. 21A), is configured to autonomously recharge its internal power source by directly interfacing with a stationary charging station (interface 29-rec at location 29 shown in FIG. 6D or a similar interface located on ground or elevated above ground). Upon navigating to the charging location (29), the vehicle (10) utilizes its sensors (4, 4-sense) to precisely locate and orient itself relative to the charging interface (29-rec). The control sub-system (8, FIG. 26) then commands the respective powered actuation system to manipulate the position and/or orientation of the main body structure (e.g., 3 in FIG. 1A; 3-body in FIG. 21A). This manipulation serves to align a designated vehicle-mounted connector with the charging interface (29-rec). While this connector could be the under-body connector (5-cnct as shown in FIG. 1A or 21A) if the charging interface is appropriately positioned (e.g., on the ground), it could alternatively be a dedicated charging port located elsewhere on the main body structure (e.g., side, front, or rear). The actuation—whether primarily vertical adjustment via linear actuators (2'+2) or multi-axis positioning via articulated legs and joints—achieves the fine alignment necessary to establish secure physical and electrical contact between the vehicle's connector and the stationary charging source (29-rec). Once charging is complete, the control sub-system (8) commands the powered actuation system to perform the reverse maneuver, disengaging the vehicle (10) from the charging station.

[0339] In this description, numerous specific details are set forth. However, the embodiments/cases of the invention may be practiced without some of these specific details. In other instances, well-known hardware, materials, structures and techniques have not been shown in detail in order not to obscure the understanding of this description. In this description, references to “one embodiment” and “one case” mean that the feature being referred to may be included in at least one embodiment/case of the invention. Moreover, separate references to “one embodiment”, “some embodiments”, “one case”, or “some cases” in this description do not necessarily refer to the same embodiment/case. Illustrated embodiments/cases are not mutually exclusive, unless so stated and except as will be readily apparent to those of ordinary skill in the art. Thus, the invention may include any variety of combinations and/or integrations of the features of the embodiments/cases described herein. Also herein, flow diagrams illustrate non-limiting embodiment/case examples of the methods, and block diagrams illustrate non-limiting embodiment/case examples of the devices. Some operations in the flow diagrams may be described with reference to the embodiments/cases illustrated by the block diagrams. However, the methods of the flow diagrams could be performed by embodiments/cases of the invention other than those discussed with reference to the block diagrams, and embodiments/cases discussed with reference to the block diagrams could perform operations different from those discussed with reference to the flow diagrams. Moreover, although the flow diagrams may depict serial operations, certain embodiments/cases could perform certain operations in parallel and/or in different orders from those depicted. Moreover, the use of repeated reference numerals and/or letters in the text and/or drawings is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments/cases and/or configurations discussed. Furthermore, methods and mechanisms of the embodiments/cases will sometimes be described in singular form for clarity. However, some embodiments/cases may include multiple iterations of a method or multiple instantiations of a mechanism unless noted otherwise. For example, when a controller or an interface are disclosed in an embodiment/case, the scope of the embodiment/case is intended to also cover the use of multiple controllers or interfaces.

[0340] Certain features of the embodiments/cases, which may have been, for clarity, described in

the context of separate embodiments/cases, may also be provided in various combinations in a single embodiment/case. Conversely, various features of the embodiments/cases, which may have been, for brevity, described in the context of a single embodiment/case, may also be provided separately or in any suitable sub-combination. The embodiments/cases are not limited in their applications to the details of the order or sequence of steps of operation of methods, or to details of implementation of devices, set in the description, drawings, or examples. In addition, individual blocks illustrated in the figures may be functional in nature and do not necessarily correspond to discrete hardware elements. While the methods disclosed herein have been described and shown with reference to particular steps performed in a particular order, it is understood that these steps may be combined, sub-divided, or reordered to form an equivalent method without departing from the teachings of the embodiments/cases. Accordingly, unless specifically indicated herein, the order and grouping of the steps is not a limitation of the embodiments/cases. Embodiments/cases described in conjunction with specific examples are presented by way of example, and not limitation. Moreover, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and scope of the appended claims and their equivalents.

Claims

1. An autonomous vehicle, comprising: a main body structure; a plurality of wheels; a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure; and a grabbing mechanism coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load beneath the main body structure and above ground; wherein the autonomous vehicle is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load relative to the ground.
2. The autonomous vehicle of claim 1, wherein the load comprises a consumer item to be delivered to a drop-off location.
3. The autonomous vehicle of claim 2, wherein the consumer item is contained within a disposable package, and wherein the grabbing mechanism is configured to directly hold said disposable package.
4. The autonomous vehicle of claim 2, wherein dimensions of the autonomous vehicle are configured to allow the vehicle to traverse a sidewalk prior to reaching the drop-off location.
5. The autonomous vehicle of claim 4, wherein said dimensions comprise a length of less than 2.5 meters, a height of less than 1.5 meters, and a weight of less than 700 kilograms.
6. The autonomous vehicle of claim 5, wherein the vehicle is configured to traverse roads prior to said sidewalk traversal.
7. The autonomous vehicle of claim 4, wherein said dimensions comprise a length of less than 1.5 meters, a height of less than 1 meter, and a weight of less than 200 kilograms.
8. The autonomous vehicle of claim 1, wherein said manipulation of the load relative to the ground comprises the vehicle being configured to: control the plurality of powered actuation systems to coordinately modify said relative positioning to lower the load towards the ground surface; and operate the grabbing mechanism to release the load onto the ground surface.
9. The autonomous vehicle of claim 8, further configured to subsequent to the grabbing mechanism releasing the load, straddle off the load.

- 10.** The autonomous vehicle of claim 1, wherein said manipulation of the load relative to the ground comprises the vehicle being configured to: position the main body structure over the load at a pick-up location; operate the grabbing mechanism to hold the load; and subsequently control the plurality of powered actuation systems to coordinately modify said relative positioning to lift the load from the ground surface.
 - 11.** The autonomous vehicle of claim 1, wherein said coordinated modification of said relative positioning, enabled by the control of the plurality of powered actuation systems to manipulate the load relative to the ground, includes maintaining a clearance between a bottom surface of the load and a ground surface during transport of the load by the autonomous vehicle.
 - 12.** The autonomous vehicle of claim 11, in which said clearance at least 10 (ten) centimeters.
 - 13.** The autonomous vehicle of claim 1, wherein said providing active suspension comprises each respective powered actuation system functioning as at least one of a spring and a damper between its associated wheel and the main body structure.
 - 14.** The autonomous vehicle of claim 1, wherein said providing active suspension enables vertical movement of each said wheel relative to the main body structure responsive to variations in the ground surface encountered during said forward motion.
 - 15.** The autonomous vehicle of claim 1, further comprising a plurality of sensors configured to collect data representing a surrounding environment of the autonomous vehicle.
 - 16.** The autonomous vehicle of claim 15, wherein the autonomous vehicle is further configured to generate, based on the data collected by the plurality of sensors, a real-time three-dimensional representation of the surrounding environment, said representation including the load when present within the surrounding environment.
 - 17.** The autonomous vehicle of claim 16, wherein said configuration to control the plurality of powered actuation systems to manipulate the load relative to the ground is based at least in part on said three-dimensional representation of the surrounding environment.
 - 18.** A method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; upon arrival at the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated modification of said relative positioning between each of the wheels and the main body structure to manipulate the load relative to a ground surface; and operating the grabbing mechanism to release the load.
 - 19.** A method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; and coordinately controlling, via the control system during at least a portion of said transporting, said plurality of powered actuation systems for maintaining a clearance between a bottom surface of the load and a ground surface.
 - 20.** The method of claim 19, in which said clearance at least 10 (ten) centimeters.
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